IntelQCopter

Autonomous Quadcopter

TECHNICAL DESIGN AND VALIDATION REPORT



AI CONCEPT ILLUSTRATION / RED AND WHITE F450 / NOT A HARDWARE TEST PHOTO

F450	1 kHz	RP2350	ROS 2
X FRAME	FC TARGET	FLIGHT CONTROLLER	PI / JETSON TBD

RP2350 aircraft + RP2040 handheld / LoRa + reserved CC1101 links.

ROS 2 onboard computer: seeking a Raspberry Pi or NVIDIA Jetson.

Models, sensor fusion and navigation design; flight acceptance pending.

IQC-PPP-002 / 28 OCTOBER 2026 / UNIVERSITY SUBMISSION DRAFT

Student / ID / institution / supervisor / academic year: complete before submission.

Proposed targets and historical results are distinguished. Motor outputs remain disabled.

github.com/BenMohStem/IntelQCopter

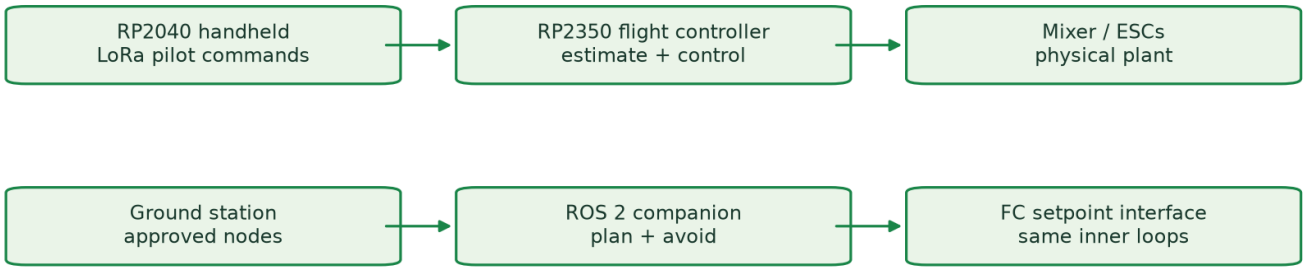
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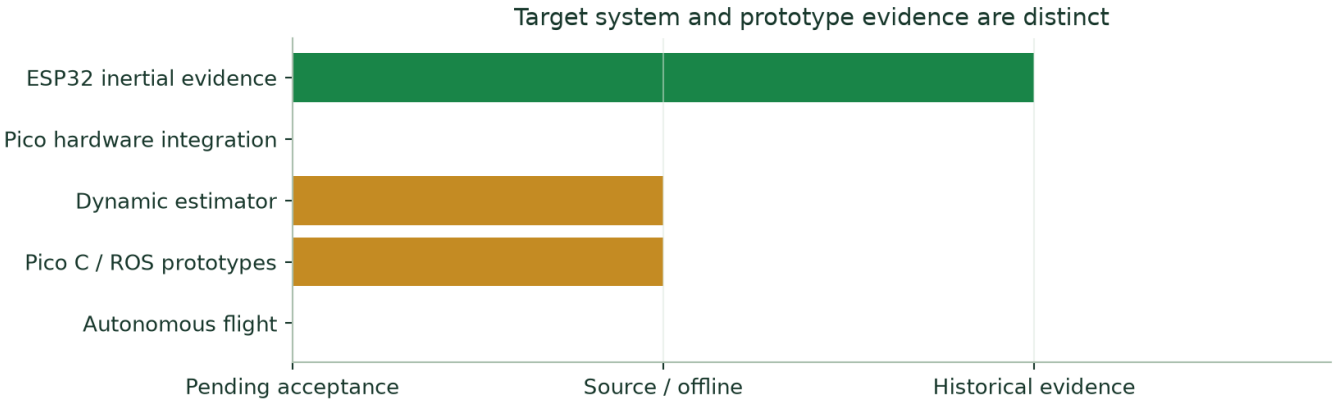
Figure label	Interpretation
Real log / historical fit	Derived from recorded project evidence; reference limits remain
Existing synthetic model	A prior numerical experiment, not flight or Gazebo
Analytic example / schematic	Explains a law or geometry under stated assumptions
Proposed architecture / campaign	Intended design or test, not a completed capability

Project in one page

Objective: a standalone stable-flight controller, extended progressively to autonomous waypoint navigation and obstacle avoidance. The aircraft must move through the actual FC command path.



PROPOSED ARCHITECTURE — Two command paths; motor commands belong only to the FC. The companion is optional.



PROJECT STATUS SUMMARY — ESP32 measurements are preliminary evidence, not RP2350 timing or flight acceptance. Pico radio/sensor integration remains pending.

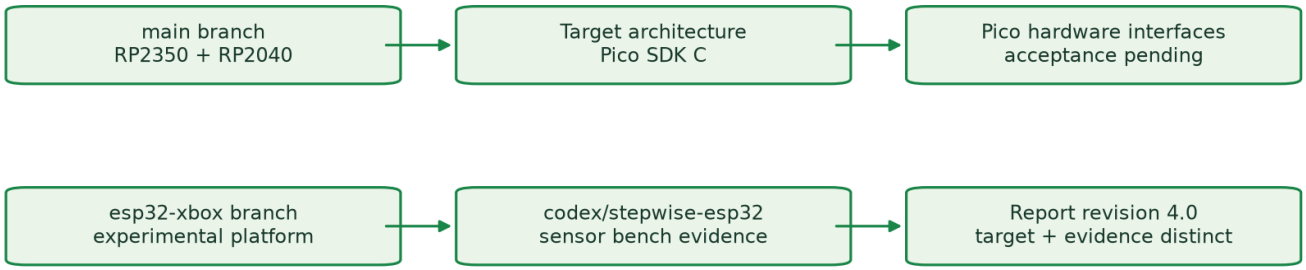
D = user decision; **P** = proposed target; **TBD** = unresolved. Historical hardware evidence, offline replay, synthetic models and planned architectures are labelled separately. No new hardware or flight experiment was run for this edition.

Target versus prototype: RP2350/RP2040 is the intended project architecture. The ESP32-S3/Xbox work is an experimental branch that supplied preliminary sensor and software evidence; those measurements are not RP2350 timing or flight acceptance. The companion is optional for stable manual flight and required for the intended onboard navigation mode.

Reading map: physics -> calibration -> estimation -> control -> perception/navigation -> verification. Complete requirements and materials follow the visual chapters. Pin wiring and low-level syntax are outside the main academic narrative; the preserved engineering baseline remains available as a reference.

Repository versions and engineering history

Public engineering repository
https://github.com/BenMohStem/IntelQCopter
Source, versions, report, measurements and scoped commit history.



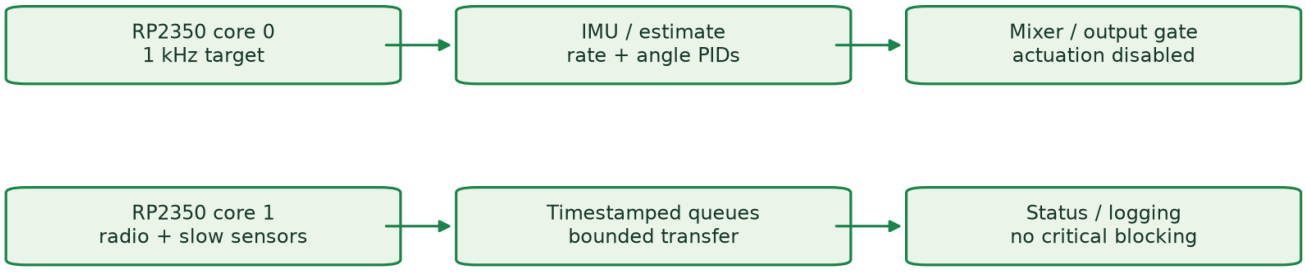
REPOSITORY VERSION MAP — Branch names were inspected locally. Different implementations and their validation results must not be conflated.

Branch or workstream	Purpose	Interpretation
main	RP2350 flight_controller.c and RP2040 handler_controller.c; Pico SDK; ROS 2 packages	Intended architecture; prototype source is not hardware/flight acceptance
esp32-xbox	ESP32-S3 / Xbox experimental implementation	Alternative prototype; do not transfer its platform-specific assumptions silently
codex/stepwise-esp32	Incremental sensor/calibration experiments and current report	ESP32 evidence retains board, source and configuration identity
ROS 2 / Gazebo	Companion, perception, mission and simulation packages	Shared research workstream; each test must name its actual engine and command path

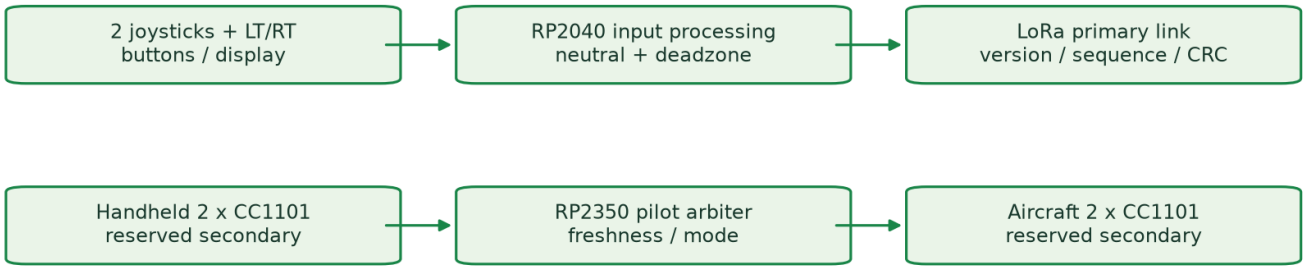
Regular version control: each meaningful completed change receives a descriptive local commit and a GitHub push after relevant checks. Preserve failed logs and earlier versions. If a step fails, repair it in a new commit and revalidate its dependants; do not rewrite history or silently reuse old acceptance. Conversations alone do not need empty commits.

Report lineage: v2.1 detailed engineering baseline -> v3 visual mathematical edition -> v4 Pico-target system edition. Earlier documents remain in reference files and Git history. The repository is public; credentials, environments and unrelated personal material are excluded. Source/code baselines and hashes are listed separately from measured results.

RP2350 flight controller and RP2040 handheld



PROPOSED ARCHITECTURE — Proposed target partition. Existing Pico scaffold is 500 Hz with stubbed hardware; RP2350 timing remains unverified.



PROPOSED ARCHITECTURE — One LoRa and two reserved CC1101 modules per side. The secondary links are not an implemented safety system.

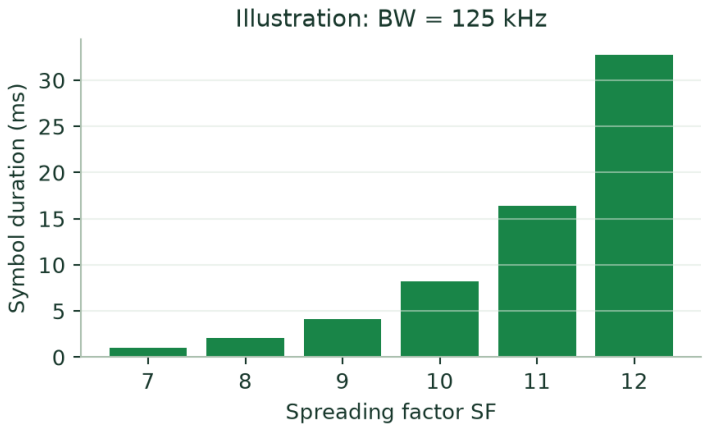
Aircraft: RP2350 Cortex-M33 target, Pico SDK, one flight_controller.c. Core 0 is proposed for the bounded 1 kHz estimation/control path; core 1 handles radio and slower work through timestamped queues. Each bus has one owner; slow requests sharing the IMU bus must be scheduled without blocking the critical path. RP2350 provides hardware single-precision floating point [R10].

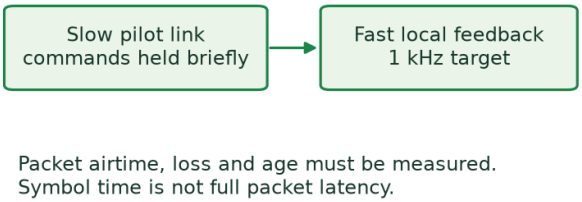
Handheld: RP2040, Pico SDK, one handler_controller.c; two centred joysticks, LB/RB, LT/RT, additional direction/mode buttons and a display. Neutral calibration, dead zone, filtering, debounce, sequence and link-health handling precede transmission. Six analogue inputs require an acquisition solution: the existing prototype uses a 74HC4051 multiplexer; settling and channel calibration need physical checks. Display, power source and final gesture mapping remain to select.

Actual source status [P6]: the Pico FC scaffold still declares a 500 Hz loop and stub radio/sensor/output interfaces; the report's 1 kHz target therefore requires implementation and fresh RP2350 profiling. Neither the existing scaffold nor ESP32 measurements prove this target. Physical outputs remain disabled.

Preliminary calibration informs the design but is not transferred blindly: changing mounting, range, scale or acquisition settings requires affected checks to be repeated.

LoRa command link and reserved CC1101 channels





Packet airtime, loss and age must be measured.
Symbol time is not full packet latency.

ANALYTIC RADIO EXAMPLE + TARGET ARCHITECTURE — The air link does not transmit a new pilot packet every 1 ms. No module/range/packet-rate acceptance is implied.

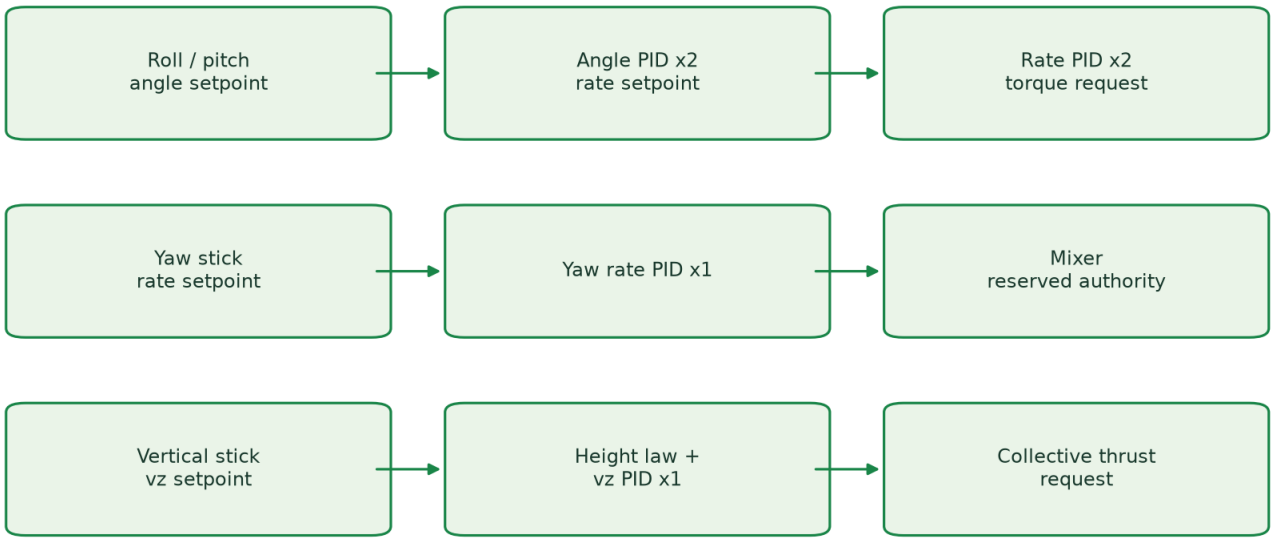
$$T_{sym} = \frac{2^{SF}}{BW}, \qquad T_{link} = T_{air} + T_{queue} + T_{decode}, \qquad T_{age} < T_{valid}$$

One LoRa module on each side carries the principal pilot-command link. Two CC1101 modules on each side are reserved for secondary data links; their roles are not implemented or assumed safety redundancy. SPI/UART details depend on the chosen LoRa module; CC1101 is an SPI-controlled sub-GHz transceiver [R11].

Interface contract: version, sequence, normalized sticks/triggers, buttons, CRC and validity age. The legacy Pico command frame is 22 bytes; future mode/mission changes require an explicit versioned extension. The LoRa cadence is selected from measured packet airtime, interference and required command age, not from the 1 kHz inner-loop rate. The FC holds bounded references between fresh packets.

Acceptance: exact modules, antennas, band, power supply, command rate, loss bursts, delay, reconnect and radio coexistence. Multiple radios sharing power, spectrum or antennas are not automatically independent. Stale commands must be rejected; the flight-phase loss response remains to approve. Local RF operating constraints must be checked before transmission.

Flight behaviour and control architecture



PROPOSED ARCHITECTURE — Six full PID blocks are requested. The outer height-to-speed law remains a design choice.

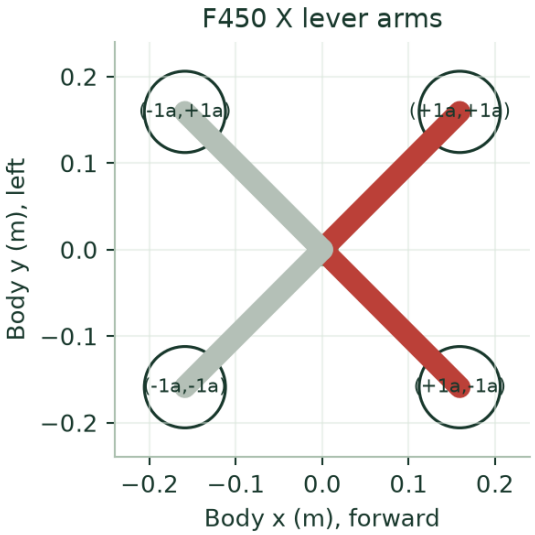
$$e_k = r_k - y_k, \qquad u_k = K_p e_k + K_i \sum_{j=0}^{k} e_j \Delta t_j + K_d \dot{e}_{f,k}$$

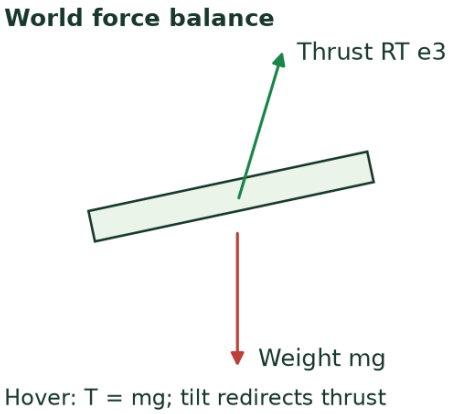
Pilot input	Controlled variable	Neutral behaviour
Right stick	Roll and pitch angles	Level attitude; no horizontal brake or position hold
Left horizontal	Yaw rate	Zero commanded yaw rate; no absolute-heading hold
Left vertical	Vertical speed	Height retention through the vertical law
RT / LT	Bounded increase / decrease	Gentle profile; reserve correction authority

Six PID blocks: two angle, two rate, one yaw rate and one vertical-speed PID. An outer proportional height-to-speed law is an option, not a frozen implementation. Mode transitions, boost interaction and vertical saturation still require design review.

Personal design rationale: I chose PID control to apply the feedback-control concepts I learned: error feedback, proportional response, integral rejection of persistent error, derivative damping, stability and sampling. Its interpretable gains let me connect mathematics to measurements. This is an educational and engineering choice, not a claim that PID is optimal for every flight condition. Cascades, filtering and saturation must still be justified experimentally.

F450 geometry and force balance





ANALYTIC GEOMETRY — Frame convention is explicit; this drawing does not assign motor numbers or spin signs.

$$L = 0.225\ \text{m}, \qquad a = \frac{L}{\sqrt{2}} \approx 0.1591\ \text{m}$$

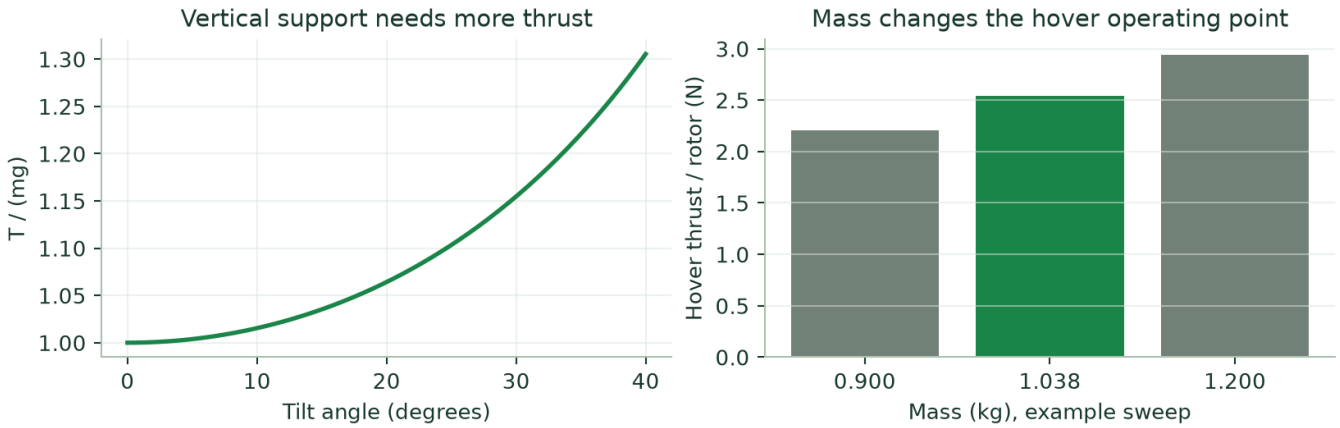
$$T = \sum_{i=1}^{4} T_i, \quad m\dot{\mathbf{v}} = R\mathbf{e}_3 T - mg\mathbf{e}_3 - D\mathbf{v} + \mathbf{f}_{ext}$$

$$J\dot{\boldsymbol{\omega}} = \boldsymbol{\tau} - \boldsymbol{\omega} \times (J\boldsymbol{\omega}), \qquad \boldsymbol{\tau}_i = \mathbf{r}_i \times \mathbf{F}_i$$

Meaning: sum the rotor forces in one reference frame; subtract gravity; the remaining force accelerates the mass. Lever arms convert each rotor force into roll/pitch torque. Rotor reaction torque contributes yaw. The cross term is angular-momentum coupling in rotating body axes.

Frame proposal: ENU world, FLU body. Confirm calibration conventions and physical motor corners before mixing. Geometry is nominal; CW/CCW signs are not assigned by this illustration.

Hover operating point and uncertain parameters



ANALYTIC EXAMPLE — At the supplied mass, hover requires about 2.545 N per rotor; actual capability is unmeasured.

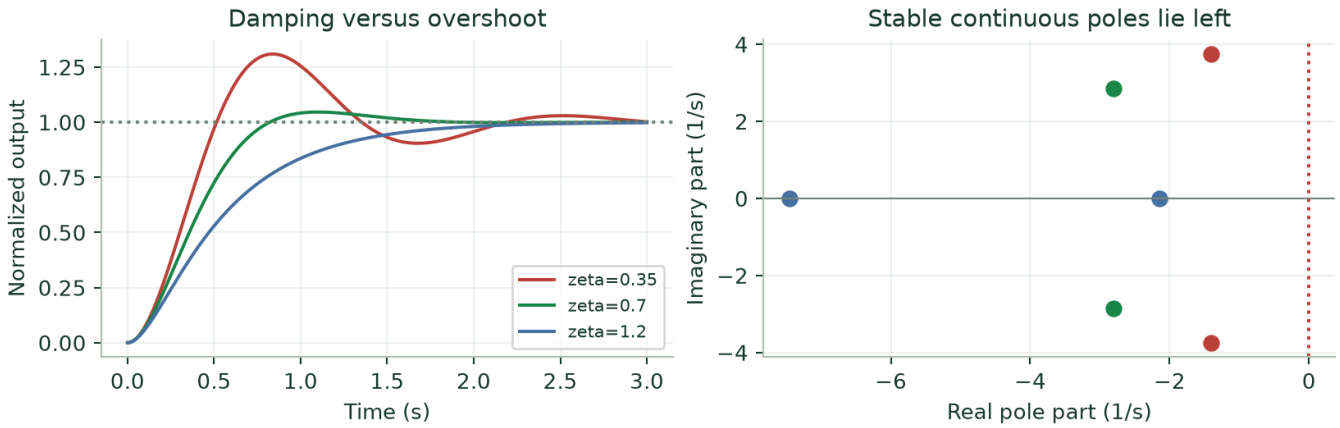
$$T_0 = mg \approx 10.179 \text{ N}, \qquad T_{i,0} \approx 2.545 \text{ N} \quad (m = 1.038 \text{ kg})$$

Parameter	Supplied model value	Qualification
Mass	1.038 kg; stated sigma 0.0160312 kg	Simulation input; verify mass and covariance assumptions
Jxx, Jyy	0.006421 kg m squared	Approximate inertia about centre of mass
Jzz	0.011853 kg m squared	Products of inertia not established
Battery	3S 4200 mAh; marked 120C	320 g in parameter file versus old rough 150 g estimate; weigh
Propulsion	1400 KV; 9047; 40 A ESCs	Exact motor and thrust/command response unknown

$$J_{xx} = \sum_i [J_{xx,i}^{COM} + m_i(y_i^2 + z_i^2)], \qquad \Sigma_y \approx G\Sigma_p G^T$$

Measurement gate: complete mass, centre of mass, motor positions, thrust curve, actuator lag, torque coefficient and power demand. KV, ESC amperage and C markings cannot supply those measurements. Propulsion tests require separate authorization and a suitable fixture.

Linear model and intuitive stability



ILLUSTRATIVE LTI MODEL — Educational second-order responses, not tuned drone gains or simulated aircraft trials.

$$G_\phi(s) = \frac{1}{J_{xx}s^2}, \quad G_\theta(s) = \frac{1}{J_{yy}s^2}, \quad G_r(s) = \frac{1}{J_{zz}s}, \quad G_z(s) = \frac{1}{ms^2}$$

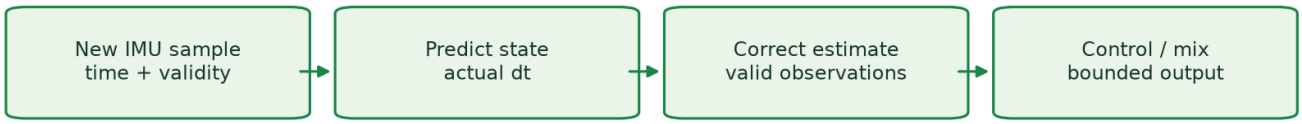
$$G_{act}(s) = \frac{1}{\tau_m s + 1}, \qquad s^2 + 2\zeta\omega_n s + \omega_n^2 = 0$$

$$K_p = J\omega_n^2, \qquad K_d = 2\zeta\omega_n J \quad \text{(ideal torque PD example)}$$

These transfer functions are local hover approximations, not global flight models. The step-response plot illustrates damping; it does not identify drone gains. Add measured actuator lag, filtering, delays, voltage effects and saturation before design acceptance.

Proposed small-signal targets: phase margin >=45 degrees; gain margin >=6 dB; overshoot <=10 percent; settling within 5 percent <=1 s. Final six-loop numerical gains remain TBD pending identification. Compare the local model with nonlinear physics rather than assuming the two are identical.

Discrete controller and sample timing

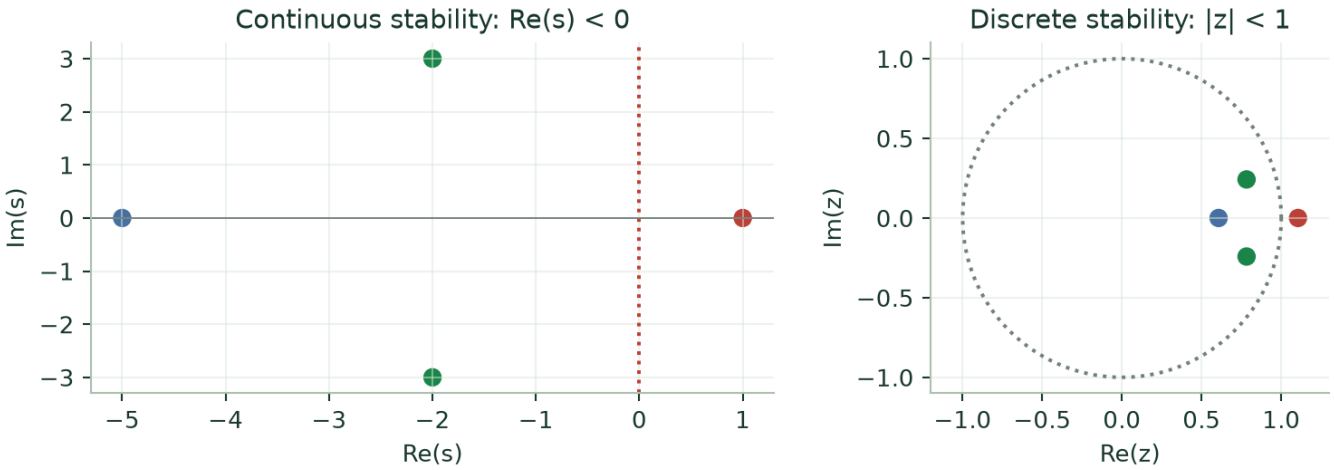


PROPOSED ARCHITECTURE — Sample, estimate, control and ESC update are distinct stages; slow observations are asynchronous.

$$\dot{\mathbf{x}} = A\mathbf{x} + B\mathbf{u} \implies \mathbf{x}_{k+1} = A_d\mathbf{x}_k + B_d\mathbf{u}_k$$

$$A_d = e^{A\Delta t}, \qquad B_d = \int_0^{\Delta t} e^{A\tau} B \, d\tau$$

$$\phi_{k+1} = \phi_k + \Delta t \, p_k + \frac{\Delta t^2}{2J_{xx}}\tau_{x,k}, \qquad p_{k+1} = p_k + \frac{\Delta t}{J_{xx}}\tau_{x,k}$$

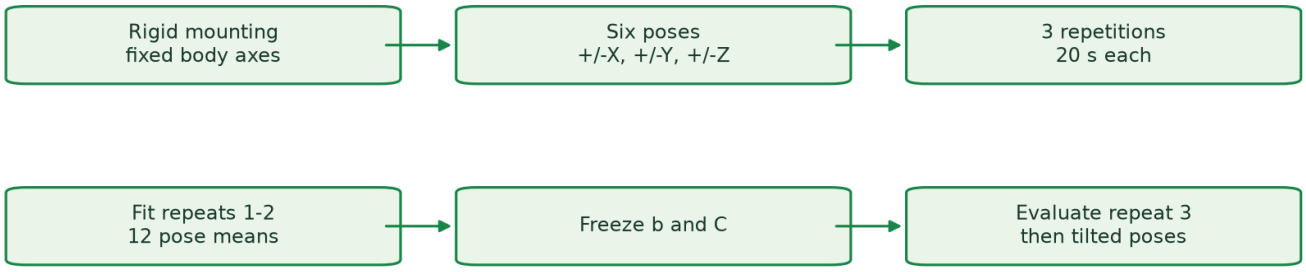


ANALYTIC SAMPLING EXAMPLE — This example uses 100 ms to show the mapping clearly; it is not the 1 ms FC implementation or its poles.

Target period: 1 ms. Timestamp acquisition, estimation, PID and output separately. Slow sensors update at their own cadence; an old sample is not new because the controller reads it again. ESC command rate depends on the actual ESC protocol, not its 40 A rating.

Implementation rule: coherent state snapshots, bounded queues/timeouts and one owner per I2C bus. A full MPU reset requires bypass restoration before accessing downstream sensors.

Calibration experimental design



HISTORICAL TEST DESIGN — Existing 18-pose design separates fitting and evaluation. Fixture uncertainty remains.

$$\mathbf{r}_j = \mathbf{b} + A\mathbf{a}_j + \boldsymbol{\varepsilon}_j, \qquad \mathbf{a}_j \in \{\pm g\mathbf{e}_x,\ \pm g\mathbf{e}_y,\ \pm g\mathbf{e}_z\}$$

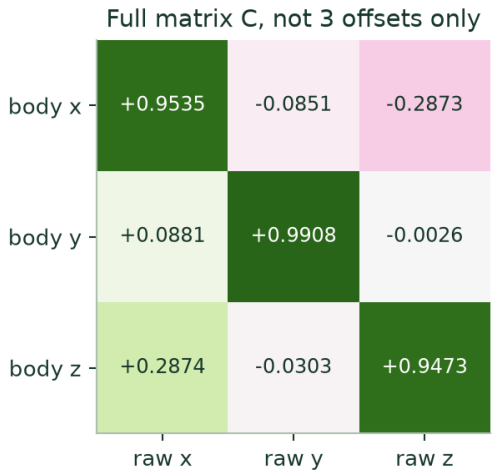
$$H = [\,\mathbf{h}_1 \cdots \mathbf{h}_{12}\,], \quad \mathbf{h}_j = [1, a_{x,j}, a_{y,j}, a_{z,j}]^T, \quad Y = [\,\mathbf{r}_1 \cdots \mathbf{r}_{12}\,]$$

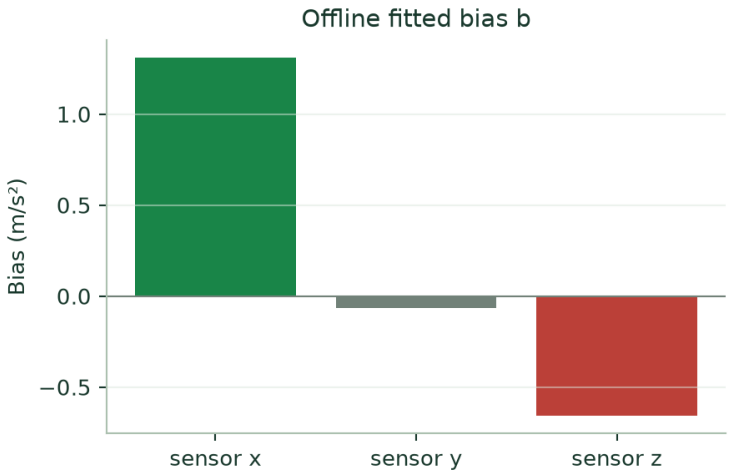
$$\widehat{M} = YH^T(HH^T)^{-1} = [\,\widehat{\mathbf{b}}\ \widehat{A}\,], \qquad C = \widehat{A}^{-1}$$

Identifiability: H must have full row rank. Six independent gravity directions reveal offset and directional response; one level pose cannot identify the full matrix. Equal fit weights do not imply exact pose references.

Evidence [P1]: 18 recordings, 20 seconds each; repetitions 1-2 fit the candidate and repetition 3 is reserved. A detached mounting invalidates its associated dataset. The secured-mount data and later tilted-pose review remain distinct. Standard gravity 9.80665 m/s squared is an assumption, not a local gravimeter reading.

Full matrix correction and held out residuals





HISTORICAL OFFLINE FIT — Coefficients shown are the historical fit candidate, not a new calibration or an instruction to overwrite trims.

$$\mathbf{a}_{cor} = C(\mathbf{r} - \mathbf{b})$$



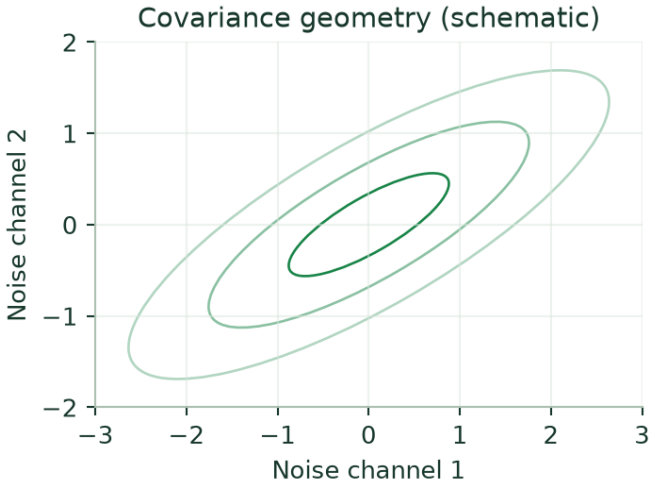


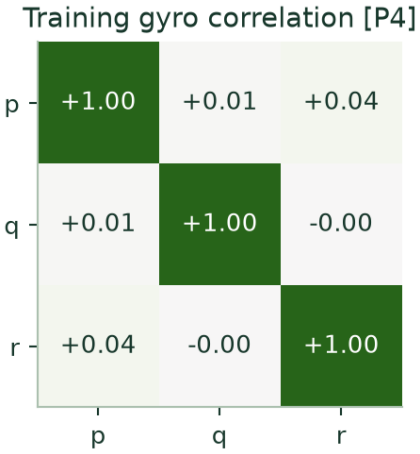
REAL LOG REPLAY SUMMARY — These residuals include reference/placement error; they are not certified sensor accuracy.

Read the heatmap by row: each corrected body axis may depend on all three raw axes. C includes scale/cross-axis/mounting effects in this historical fit. The bias and matrix are not interchangeable with three MPU offset registers.

Do not recalibrate from this page. These are historical offline-fit coefficients [P1], not a newly installed calibration. Residuals contain fixture/placement uncertainty; they are not certified sensor accuracy. Frozen firmware constants require their own recorded provenance and reset/power-cycle verification.

Covariance and useful information

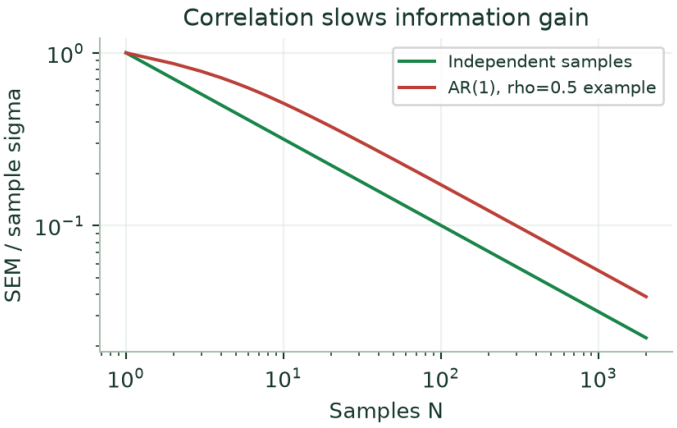


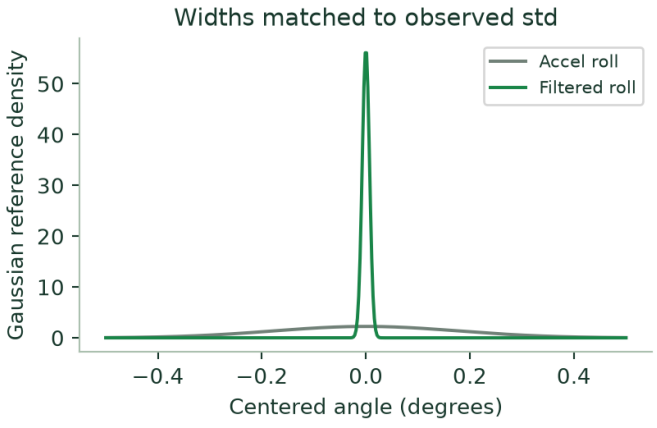


SCHEMATIC + MEASURED SUMMARY — Off-diagonal terms show shared variation. Zero covariance does not generally imply independence.

$$\bar{\mathbf{x}} = \frac{1}{N}\sum_i \mathbf{x}_i, \qquad S = \frac{1}{N-1}\sum_i (\mathbf{x}_i - \bar{\mathbf{x}})(\mathbf{x}_i - \bar{\mathbf{x}})^T$$

$$\mathrm{Cov}(\bar{\mathbf{x}}) = \frac{1}{N^2}\sum_i \sum_j \mathrm{Cov}(\mathbf{x}_i, \mathbf{x}_j)$$

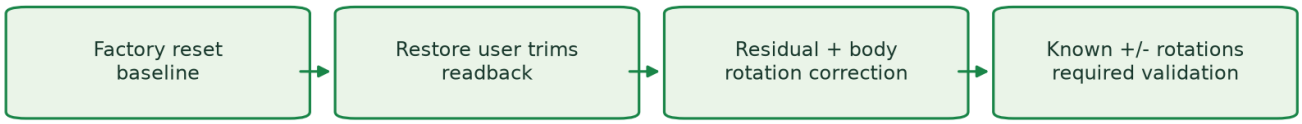




ANALYTIC EXAMPLE + OBSERVED WIDTHS — Density shapes are explanatory references, not a claim that the actual noise is Gaussian or unbiased.

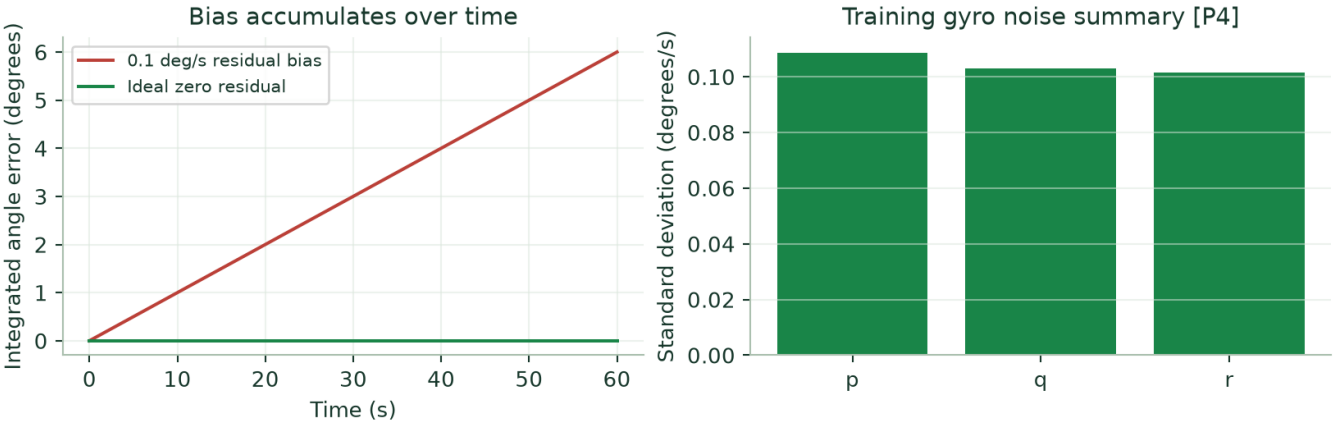
For independent samples, the last expression reduces to Sigma/N. With correlation, cross-time terms remain. More samples do not automatically remove drift, reference error or modelling bias. The density curves show widths only; neither joint Gaussianity nor independence is asserted. Report repeatability and accuracy separately.

Gyro calibration and frame validation



PROPOSED ARCHITECTURE — Hardware bias correction and software transform have separate roles; gyro scale/physical axes remain pending.

$$\boldsymbol{\omega}_b = R_{bs}(\boldsymbol{\omega}_s - \mathbf{b}_g), \qquad \Delta\theta_{bias}(t) = b_g t$$

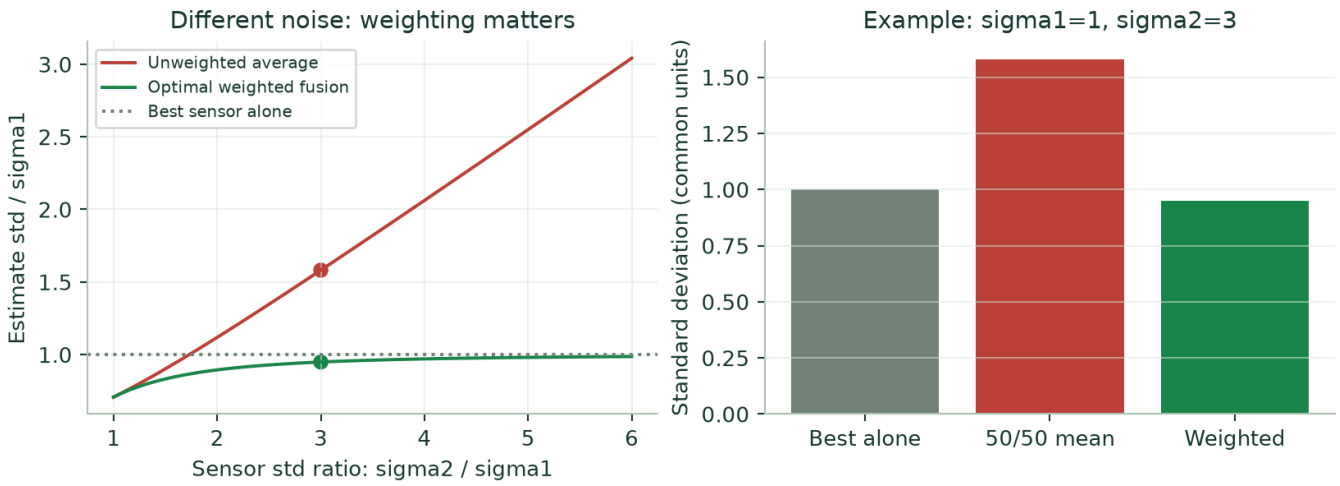


ANALYTIC EXAMPLE + MEASURED SUMMARY — Averaging reduces random fluctuation but does not remove an unmodelled residual bias.

Bias is not scale or frame mapping. Motionless data estimate bias/repeatability. Known positive and negative rotations about physical body X, Y and Z test sign and mapping; a measured rotation reference is additionally needed for scale.

MPU user offsets are volatile. Restore against the documented reset baseline, read back, then apply only the remaining software bias. Do not subtract the full bias twice or use the accelerometer scale matrix as a gyro rotation. Physical gyro direction/scale checks remain pending.

Multisensor fusion versus a simple average



ANALYTIC INDEPENDENT-SENSOR EXAMPLE — Inverse-variance fusion improves the model variance. Correlation, bias, different units or wrong dynamics invalidate the simple comparison.

For two unbiased independent measurements of the same scalar x, let y1 = x + e1 and y2 = x + e2, with variances R1 and R2. Choose an unbiased linear estimate xhat = (1-w)y1 + w y2; its variance is minimized by inverse-variance weighting.

$$P(w) = (1-w)^2 R_1 + w^2 R_2, \quad \frac{dP}{dw} = 0 \implies w^* = \frac{R_1}{R_1 + R_2}$$

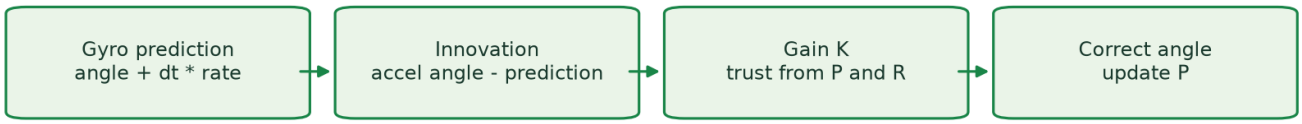
$$P^* = \frac{R_1 R_2}{R_1 + R_2} \le \min(R_1, R_2), \qquad P_{average} = \frac{R_1 + R_2}{4}$$

$$\mathbf{w}^* = \frac{R^{-1}\mathbf{1}}{\mathbf{1}^T R^{-1}\mathbf{1}}, \qquad P^* = (\mathbf{1}^T R^{-1}\mathbf{1})^{-1} \quad \text{(full covariance)}$$

Example: sigma1 = 1 and sigma2 = 3 in the same units. A 50/50 average gives sigma = 1.581, worse than sensor 1 alone. Optimal weighting gives sigma = 0.949. The noisier sensor contributes only 10 percent. No Gaussian assumption is required for this minimum-variance linear weighting under the stated covariance model.

Kalman connection: a prediction is fused with an observation using uncertainty and dynamics. Gyro rate and acceleration cannot be averaged directly: their units and models differ. Integrate the gyro and derive a valid gravity-angle observation first. With equal independent static-sensor variance, the ordinary mean is already optimal among unbiased linear combinations. Known correlation requires the full positive-definite covariance R; bias or model mismatch can invalidate the predicted physical improvement.

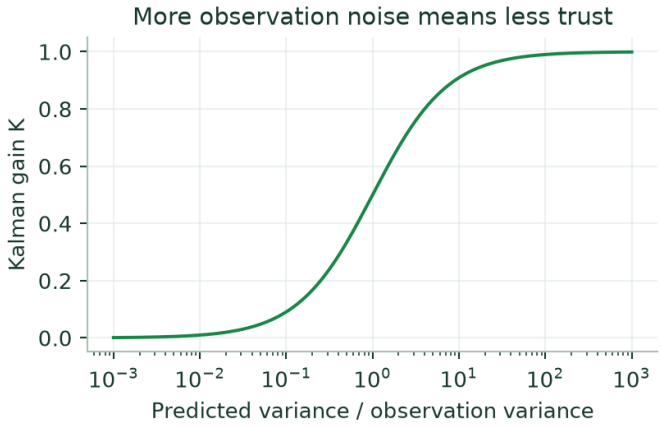
Scalar Kalman filter and trust

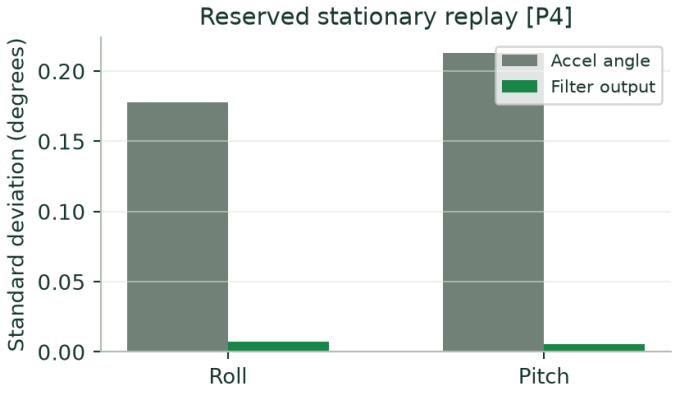


PROPOSED ARCHITECTURE — Scalar roll/pitch estimator candidate. Translational acceleration can corrupt the gravity observation.

$$\theta_k^- = \theta_{k-1} + \Delta t\,\omega_k, \quad P_k^- = P_{k-1} + Q_k, \quad \nu_k = z_k - \theta_k^-$$

$$K_k = \frac{P_k^-}{P_k^- + R_k}, \qquad \theta_k = \theta_k^- + K_k\nu_k, \qquad P_k = (1 - K_k)P_k^-$$



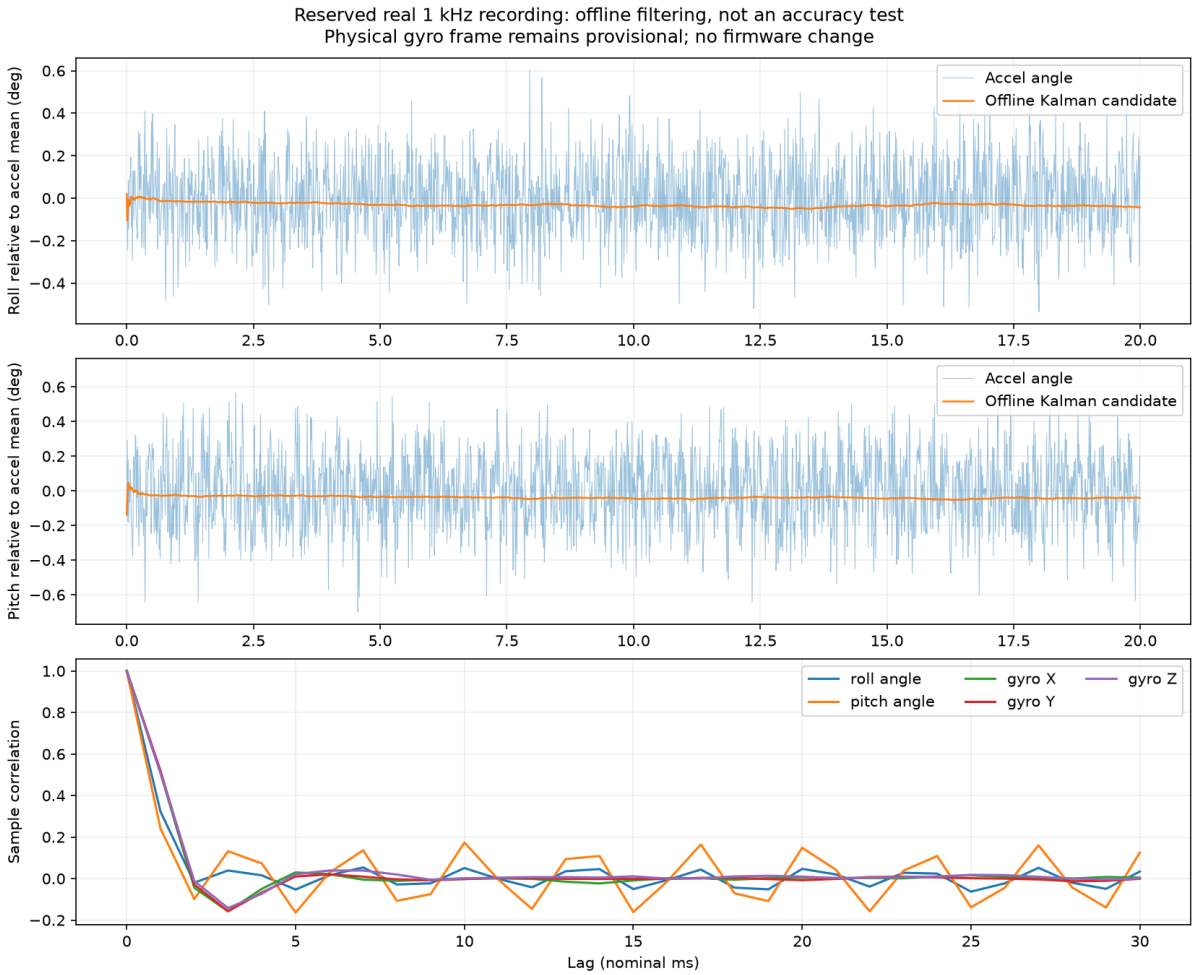


ANALYTIC LAW + REAL REPLAY SUMMARY — Smoother output is not an independent accuracy result; mean errors and correlation remain.

Historical scalar model [P4]	Q (rad squared / step)	R (rad squared)	Conditional limiting K
Roll	3.5948594e-12	9.6143606e-6	0.00061129
Pitch	3.2286779e-12	1.3818086e-5	0.00048326

For independent discrete gyro-rate noise, angle-process variance is sigma_rate squared times dt squared. Continuous white-rate spectral density instead integrates proportionally to dt. Keep those models and units distinct. Existing candidate gains are model-dependent; stationarity and autocorrelation limit their interpretation [P4].

Original recording replay

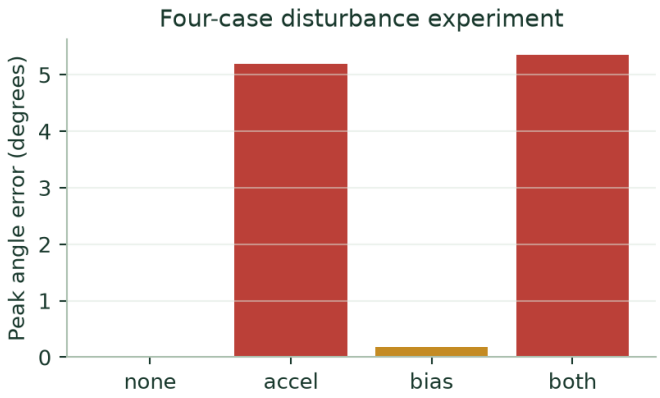


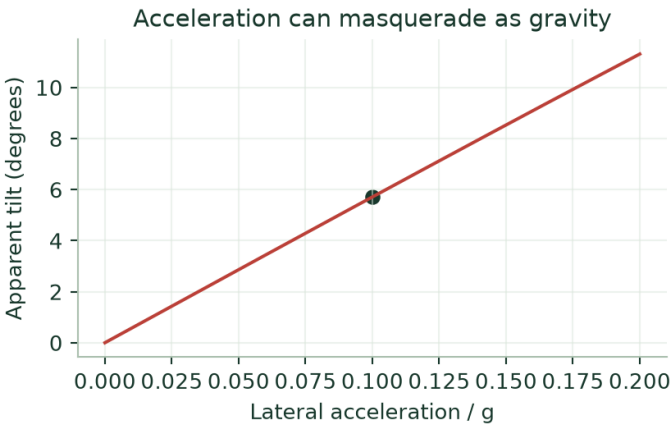
Original derived figure [P4], reproduced without alteration. Engine: Python float64 replay of a reserved real 1 kHz recording. This is not live sensor output, an MCU filter execution or a flight. All samples enter the analysis; displayed points are reduced for readability.

Reserved result	Roll (degrees)	Pitch (degrees)
Acceleration-angle standard deviation	0.177569	0.213074
Filter-output standard deviation	0.007000	0.005692
Filter mean minus observation mean	-0.033769	-0.039908

The evaluation excludes a fixed two-second initialization interval. No independently measured attitude reference exists for this recording; lower fluctuation does not demonstrate absolute accuracy. Mean differences were retained, not fitted away. The lower panel shows why independent-white-noise assumptions need checking.

Why dynamic acceleration defeats a quiet filter





EXISTING SYNTHETIC MODEL + ANALYTIC LAW — Python model results, not Gazebo or flight. The candidate is not accepted for dynamic operation.

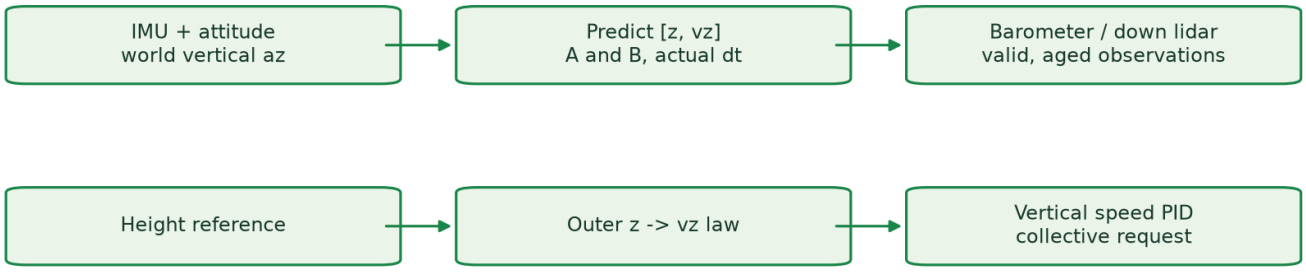
$$\theta_{apparent} = \arctan(a_{lat}/g), \qquad \|\mathbf{a}\|/g = \sqrt{1 + (a_{lat}/g)^2}$$

At a_lat = 0.1g, apparent tilt is 5.71 degrees while the norm is only about 1.005g. A magnitude-only acceptance gate can miss the disturbance.

Existing model case [P4]	RMSE (degrees)	Peak error (degrees)
No added disturbance	0.004312	0.012507
0.1g acceleration	2.114109	5.191244
0.1 degree/s gyro bias	0.157612	0.172872
Both	2.214463	5.354520

Decision: evaluate innovations, bias dynamics and acceleration-sensitive correction on independent dynamic references before acceptance. These are existing Python synthetic-model results; no Gazebo or physical flight is implied. Smoothing or collecting more stationary samples cannot remove an unobservable force component.

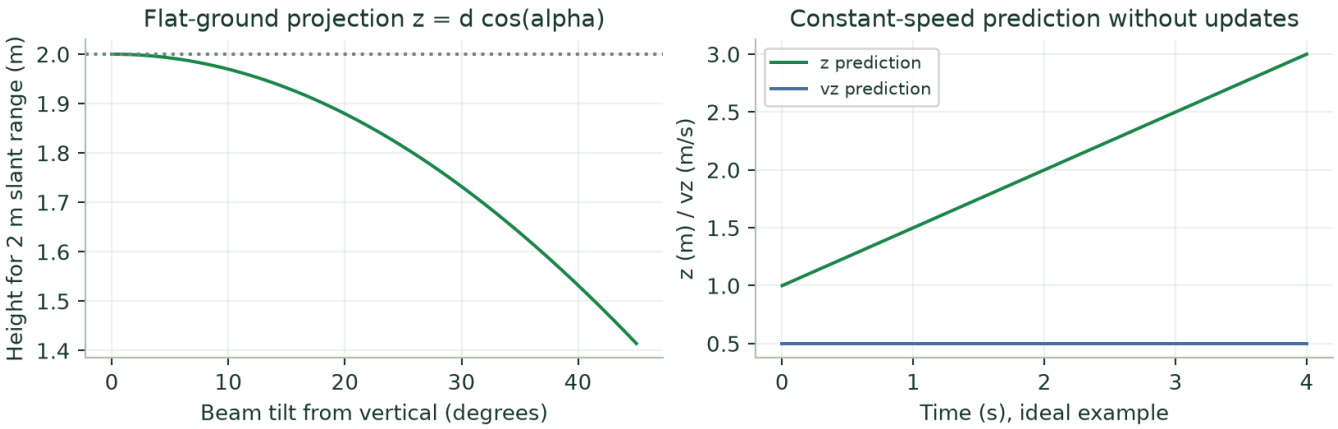
Vertical estimation and sensor geometry



PROPOSED ARCHITECTURE — Two-state estimator and vertical control architecture proposed; barometer and lidar integration remain pending.

$$z_{k+1} = z_k + \Delta t\, v_{z,k} + \tfrac{1}{2}\Delta t^2 a_{z,k}, \qquad v_{z,k+1} = v_{z,k} + \Delta t\, a_{z,k}$$

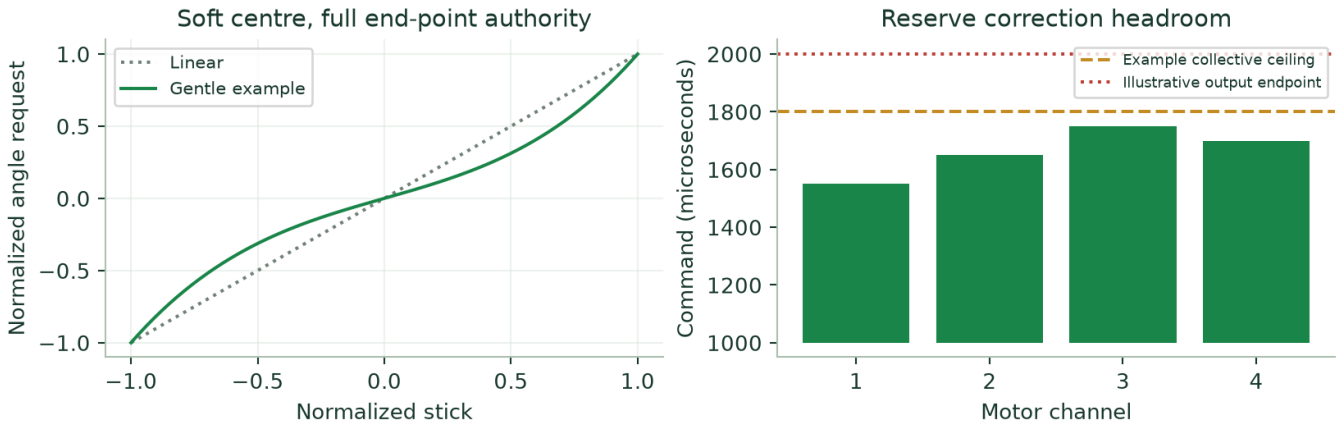
$$P_k^- = A_d P_{k-1} A_d^T + Q_k, \quad K_k = P_k^- H^T (H P_k^- H^T + R_k)^{-1}$$



ANALYTIC EXAMPLE — A forward lidar cannot supply this height observation. Terrain and extrinsics must be checked.

The two-state model estimates z and vz; it does not automatically estimate accelerometer bias. Q must follow the selected disturbance model and sample interval. A mass variance cannot be copied directly into Q. Barometer drift, lidar status, tilt projection, terrain, measurement age and dropout all require tests. Bottom-versus-front lidar placement remains a decision, not an interchangeable software setting.

Gentle control authority and anti windup



ILLUSTRATIVE CONTROL DESIGN — Actual collective/vertical law and ESC limits are TBD. A boost request is not four identical maximum outputs.

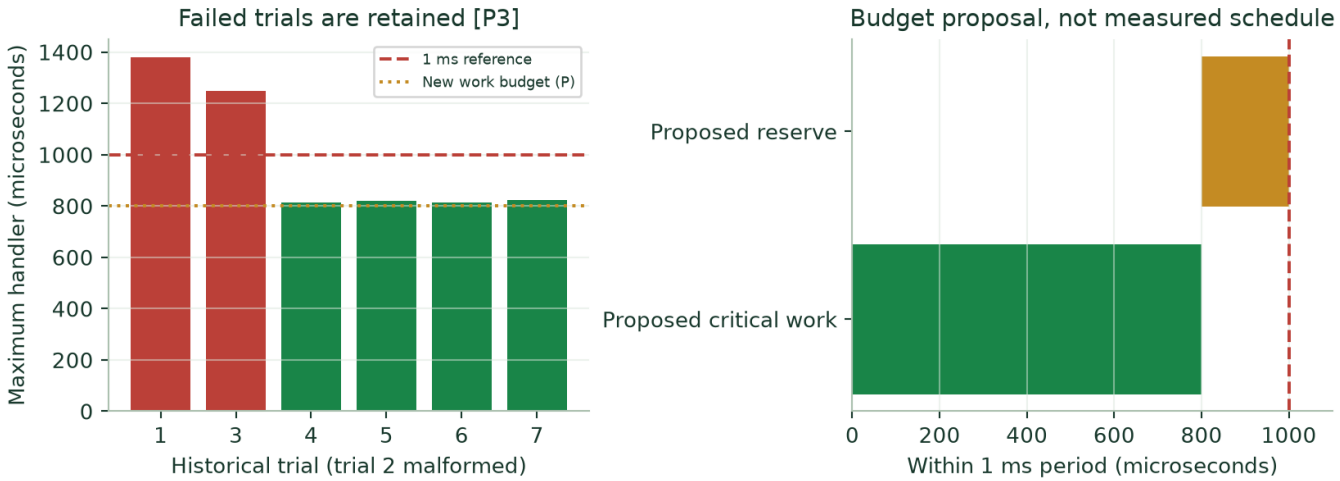
$$u_{raw} = u_P + u_I + u_D, \qquad u = \mathrm{clip}(u_{raw}, u_{min}, u_{max})$$

$$\dot{u}_I = K_i e + K_{aw}(u - u_{raw}) \quad \text{(one antiwindup option)}$$

Design priorities: bounded angle/rate requests, gentle centre response, filtered derivative, explicit saturation, attitude correction headroom and bumpless mode changes. The plotted command values and shaping curve are examples, not firmware settings or accepted ESC endpoints.

Boost is not maximum output. A proposed 1700-1800 microsecond collective ceiling preserves some room for mixing, but the exact relationship between RT/LT, vertical speed and collective remains unresolved. Verify signs, mixer geometry and spin directions before physical actuator work. All motor outputs remain disabled.

Real time evidence including failures



REAL LOG SUMMARY + PROPOSED BUDGET — Historical samples were not a complete PID controller; source hashes and inter-service gaps still matter.

$$T_s = 1000\ \mu s, \qquad T_{critical} + T_{reserve} \leq T_s$$

Measured history [P3]: trial 2 was malformed and rejected; earlier trials exceeded the handler budget. Repair separated acquisition/calculation from synchronous serial formatting using a bounded queue. The later 822 microsecond maximum leaves only 178 microseconds to 1 ms if compared directly; a complete future FC includes additional work.

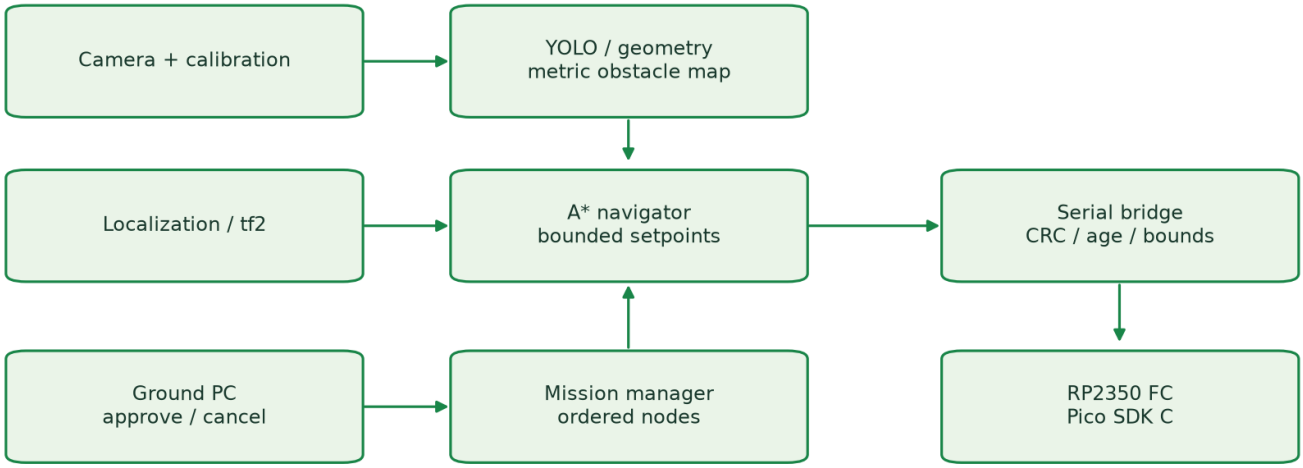
All measured timing on this page belongs to the **ESP32-S3 preliminary platform**, not RP2350. The main Pico scaffold's 500 Hz setting and unconnected drivers remain a separate implementation baseline [P6].

Proposed target: <=800 microseconds critical work, >=200 microseconds reserve, and no observed deadline miss in a defined 10-minute loaded test. Historical samples near 1000.69 Hz do not prove a periodic hard deadline. Inter-service gaps, clocks and source changes remain relevant.

P5 build only: 54 Python checks, 19 actual-source host cases, ESP32 compile; 19504 bytes RAM and 312901 bytes flash. Source f26d314e was not uploaded in that record. Historical timing source 91947b5c is different. A host fake peripheral is not an ESP32 timing measurement.

Bench source configuration: accelerometer +/-8g, gyro +/-500 degrees/s, nominal 260 Hz low-pass setting with readback checks. Sensor bandwidth is not the PID bandwidth. These settings do not validate the revised sensor wiring or a full control loop.

ROS 2 software implementation and interfaces



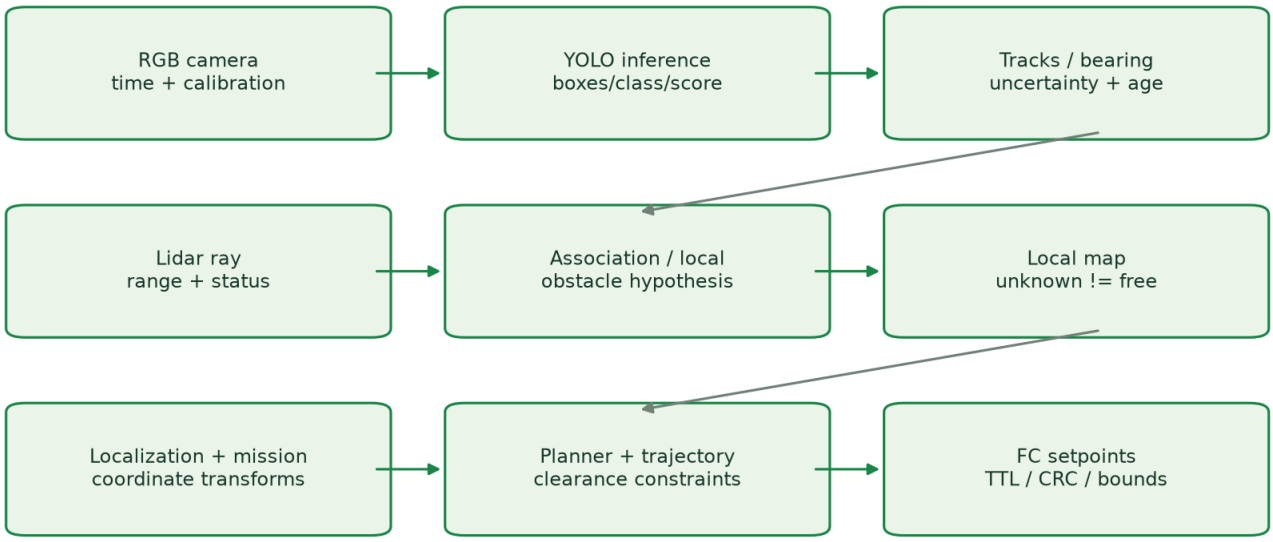
PROPOSED ROS 2 INTEGRATION — ROS 2 C++ runs on the companion. Ground-PC missions enter the mission manager; only the bounded serial interface reaches RP2350.

Boundary	Implementation contract
Ground station to companion	IP network, mission upload/approval, status and cancel; separate from LoRa pilot link
Camera and perception	Stamped RGB plus intrinsics; detections become metric obstacles only with validated geometry
Localization	Pose/velocity/covariance with coherent map, odom and body frames; freshness checked
Mission and navigator	Ordered-node progression, local-map updates, replanning and bounded references
Serial bridge to RP2350	Versioned setpoints and health; CRC, timestamps, expiry and authority arbitration
Evidence recording	rosbag2/topics, video and FC logs aligned to recorded clocks and source versions

Existing code [P6]: qcopter_autonomy contains C++ mission, navigator, YOLO and panel-pose nodes plus custom messages; qcopter_description contains the F450 URDF/RViz model. The existing mission service specifies start/finish only. A cancellable n-node ROS action with progress feedback is a proposed extension, not an already verified feature [R13].

Communication design: use small bounded histories; sensor streams prioritize freshness, mission requests need acknowledged handling, and the FC independently rejects stale commands. ROS QoS compatibility must be checked; reliable delivery alone does not guarantee a deadline [R13]. ROS 2 runs on the companion, while the RP2350 runs Pico SDK C firmware rather than a full ROS installation.

YOLO to obstacle avoidance architecture



PROPOSED ARCHITECTURE — Proposed perception-to-control pipeline. Detection is not depth, localization or a collision-free corridor.

$$\mathbf{z}_{det} = (u_1, v_1, u_2, v_2, c, s), \qquad \mathbf{d}_c \propto K_c^{-1}[u, v, 1]^T$$

$$U^* = \arg\min_U \sum_k \left(w_p \|\mathbf{p}_k - \mathbf{p}_{goal}\|^2 + w_u \|\mathbf{u}_k\|^2 + w_c \psi(d_k - d_{safe}) \right)$$

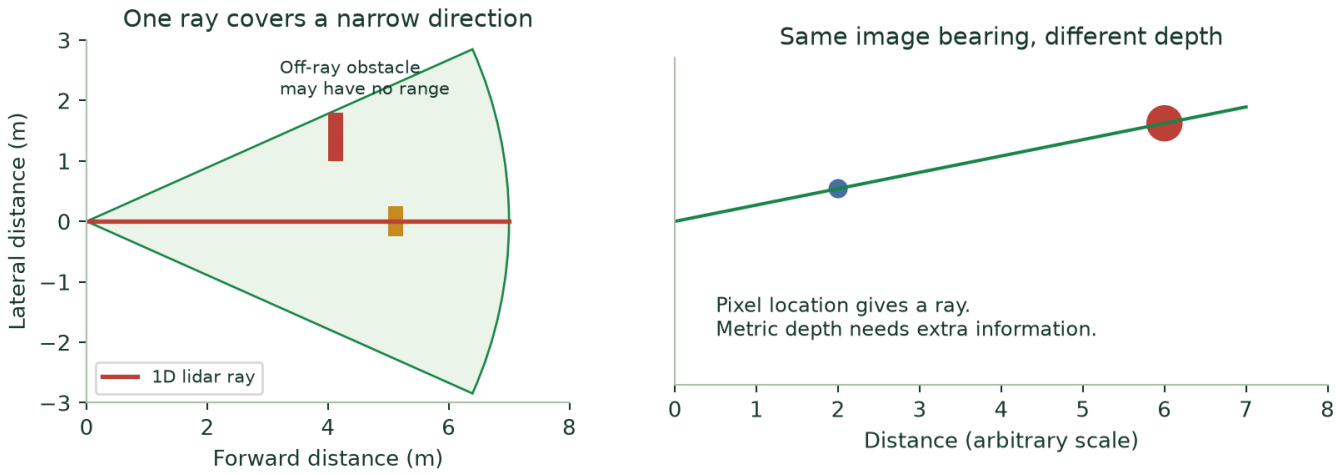
YOLO supplies: bounding boxes, class and score [R7]. It does not supply complete metric free space. Camera intrinsics Kc, camera-to-body extrinsics, timestamps, tracking and uncertainty are additional contracts.

Fusion gate: associate a lidar ray with a visual hypothesis only when geometry, timing and field of view justify it. Unassociated or unobserved space stays unknown. No detection does not imply no obstacle. Localization must make waypoint coordinates observable before a metric mission can execute.

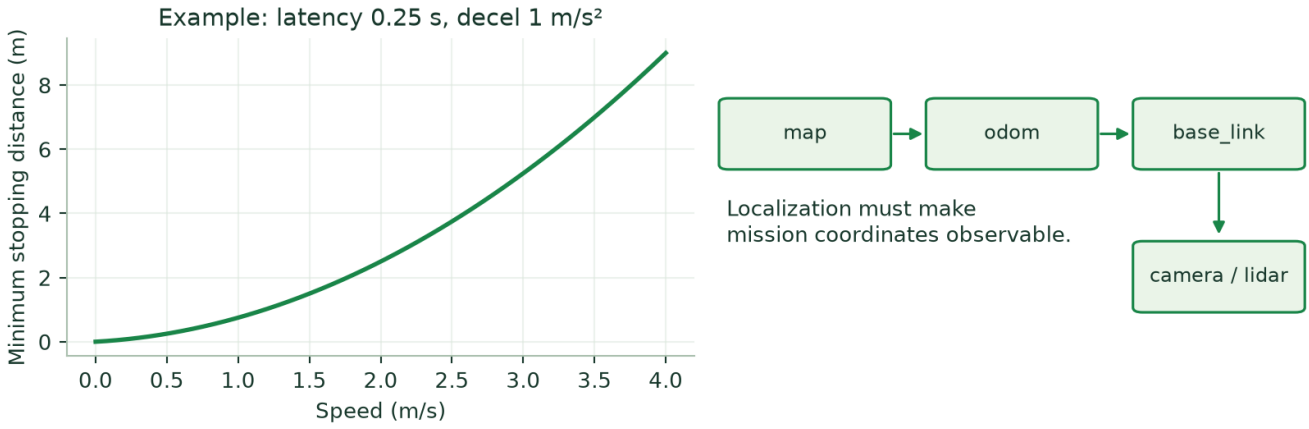
Panel obstacles may be absent from pretrained classes. Training data, panel recognition and a class-agnostic fallback remain to be selected and tested.

Candidate planning objective: penalize goal error, control effort and inadequate clearance, subject to the motion model, input bounds and observed-space constraints. U is a candidate control sequence; psi is a clearance penalty. This does not select or validate a planner. The FC remains responsible for bounded commands. Model weights, runtime, licence, speed and companion hardware remain TBD.

Perception limits and stopping clearance



GEOMETRIC SCHEMATIC — Objects, field of view and distances are illustrative, not camera specifications or YOLO detections.



ANALYTIC EXAMPLE + PROPOSED FRAMES — Real braking, latency and localization are unmeasured; speed limits cannot be inferred from this example.

$$\mathbf{p}_w = R_{wb}\left(R_{b\ell}[d, 0, 0]^T + \mathbf{t}_{b\ell}\right) + \mathbf{t}_{wb}$$

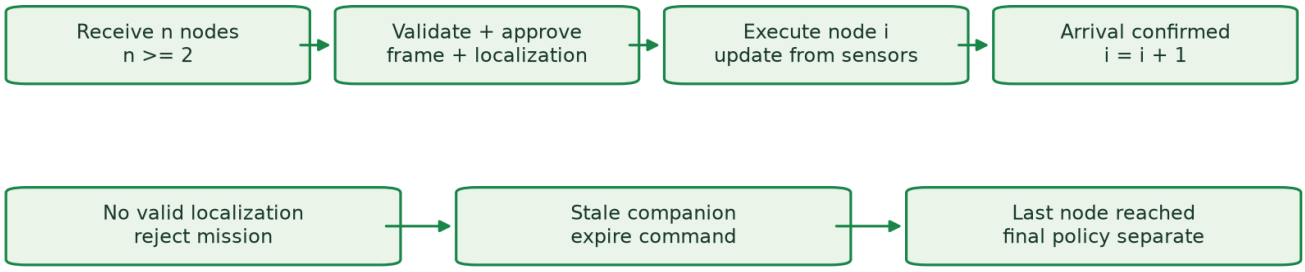
$$d_{clear} > v\tau + \frac{v^2}{2a_{brake}} + d_{margin}$$

One RGB camera plus one fixed ray cannot observe the entire collision corridor. Forward lidar helps one direction; downward lidar helps local height under terrain/tilt assumptions. Metric monocular reconstruction needs additional information and validation; a pixel bearing alone has ambiguous depth.

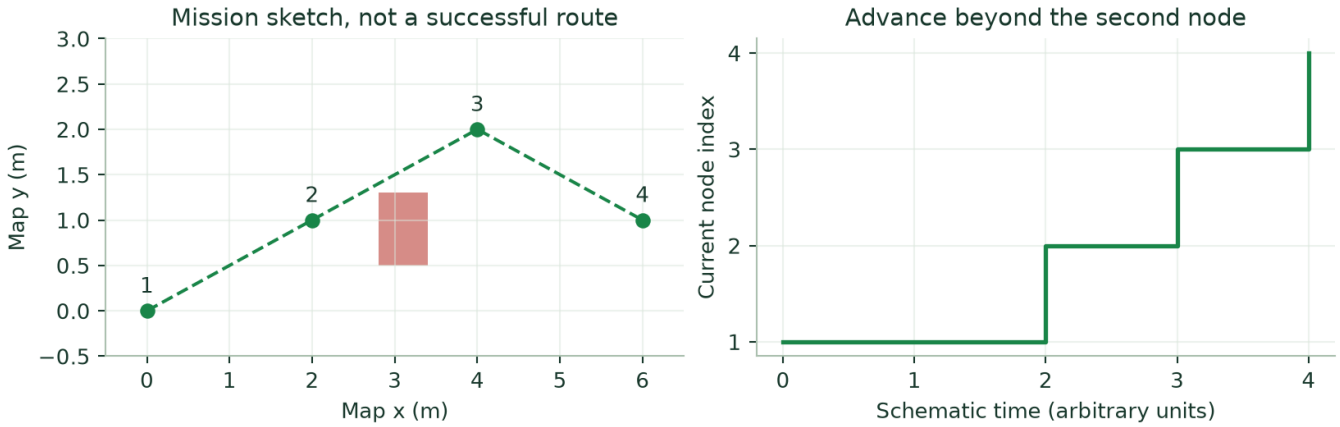
The point transform assumes the lidar ray follows its local positive x axis: rotate/translate lidar to body, then body to world. Extrinsics and pose timestamps must be known; otherwise range cannot be placed reliably in the obstacle map.

The clearance graph assumes 0.25 s total latency and 1 m/s squared braking only to explain scaling. These are not measured aircraft parameters. Reduce the operational envelope until latency, braking, localization and missed-detection behaviour are verified. Any simulator depth camera or perfect pose must be declared as extra sensing or an oracle.

Missions with two three or many nodes



PROPOSED ARCHITECTURE — Mission progression must not stop at node 2. Loss responses and final landing policy require approval.



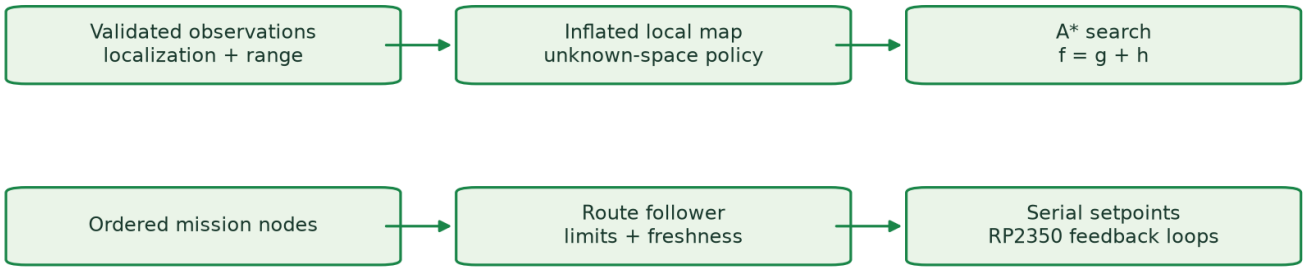
PLANNED TEST SCHEMATIC — The dashed node sequence is input geometry; it does not claim obstacle avoidance or a collision-free path.

$$\|\hat{\mathbf{p}} - \mathbf{p}_i\| \le r_{accept}, \quad \|\hat{\mathbf{v}}\| \le v_{accept} \quad \text{for} \quad t \ge t_{dwell}$$

Proposed arrival: radius 0.5 m, dwell 1 s, speed bound TBD; adapt to localization uncertainty. Increment every intermediate node, not only the first. Final arrival and landing/termination are separate policies. The node sketch is an input mission, not a computed safe trajectory.

Communication: versioned framed 3.3 V UART is proposed, with length, CRC, sequence, timestamp, units, mission identity and command expiry. Proposed heartbeat 10 Hz and expiry 300 ms require review. Invalid frames do not renew validity. Companions may be absent; plugging one in never changes mode automatically. Loss reactions remain to approve.

Navigation implementation and remaining gaps



PROPOSED ARCHITECTURE — A bounded C++ A* prototype exists; generalized n-node ROS integration, real localization and physical bridge acceptance remain pending.

$$f(n) = g(n) + h(n), \qquad h(n) = \|\mathbf{p}_n - \mathbf{p}_{goal}\|$$

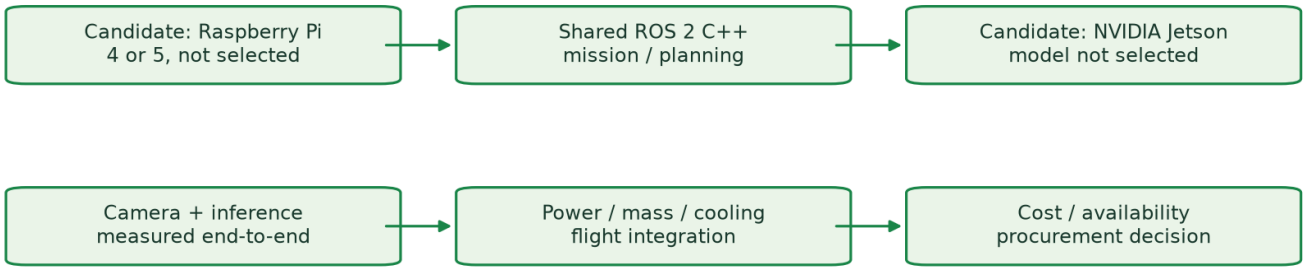
$$r_{inflate} \geq r_{vehicle} + d_{stop} + k_\sigma \sigma_{position} + d_{margin}$$

Existing algorithm [P6]: an A* grid planner and waypoint follower exist in the portable C++ core. The inspected prototype uses 0.2 m cells, fixed 0.65 m obstacle inflation and a bounded local XY map. Those constants are prototype settings, not validated aircraft clearance or general 3D navigation. The target must account for swept propeller volume, stopping distance and localization uncertainty.

Existing range demonstration: known-size blue panels can supply monocular PnP range when their 1.09 x 1.69 m inner dimensions and trained detector are explicitly confirmed. This does not solve arbitrary obstacle depth. Simulated odometry/contact and any optional depth-image branch must be declared; a real localization source and unknown-space policy remain unresolved.

Implementation sequence: synchronize sensors and transforms -> validate metric obstacle observations -> update the local map -> replan -> track the route with bounded angle/yaw-rate/vertical references -> send through the RP2350 bridge. Iterate intermediate nodes; do not teleport or inject forces. Loss of valid pose/perception must invalidate autonomous commands under an approved response policy.

Onboard computer selection remains open



PROPOSED ARCHITECTURE — The project is seeking one of these onboard computers. Common source does not imply the same OS image or acceleration backend.

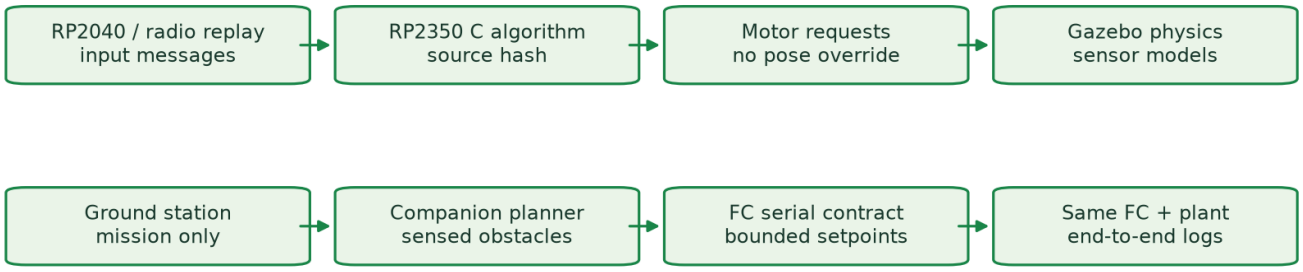
We are looking for either a Raspberry Pi 4/5 or an NVIDIA Jetson for the onboard ROS 2 computer. Neither is selected or procurement-confirmed. The FC remains independently usable without it; autonomous mission execution needs a suitable companion and valid localization.

Decision factor	Raspberry Pi candidate	Jetson candidate
Common software	Same portable ROS 2 C++ mission/navigation core	Same portable ROS 2 C++ mission/navigation core
Inference	Benchmark CPU or selected accelerator/runtime	Benchmark supported GPU/runtime for the exact Jetson
Integration	Camera driver, supported OS/ROS and serial bridge	Camera driver, supported OS/ROS and serial bridge
Flight cost	Board, storage, camera, regulator, cooling and mounts	Board, storage, camera, regulator, cooling and mounts
Selection evidence	Measured latency, power, mass, temperature and cost	Measured latency, power, mass, temperature and cost

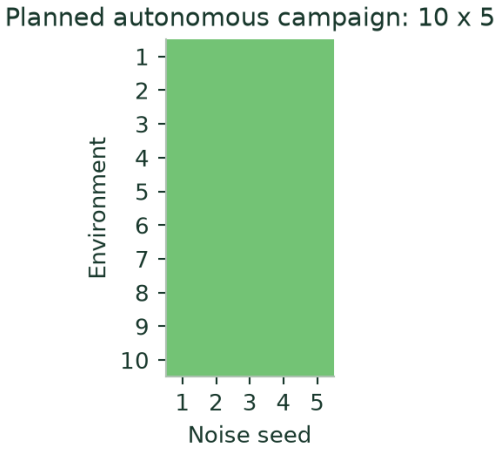
$$m_{new} = m_{baseline} + \Delta m_{FC/RF} + m_{computer} + m_{camera} + m_{power/mounts}$$

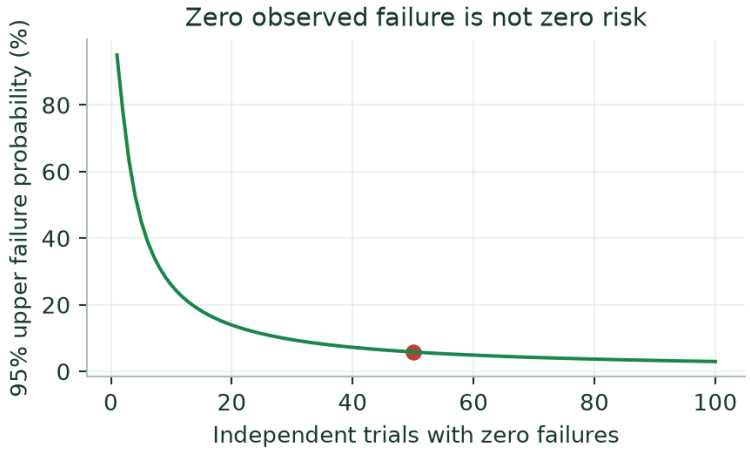
The 1.038 kg parameter set predates this target assembly. Delta m_FC/RF includes replacing the prototype controller and adding radios/antennas. Recompute mass, centre of mass and inertia. Shared C++ source does not imply identical binaries, inference backends or OS images. Select from measured end-to-end latency, localization quality, power, mass and cost, not nominal AI throughput alone.

Proving that the FC really flies the simulation



PROPOSED ARCHITECTURE — A test harness may stimulate inputs and record outputs; it may not secretly move the aircraft to pass.





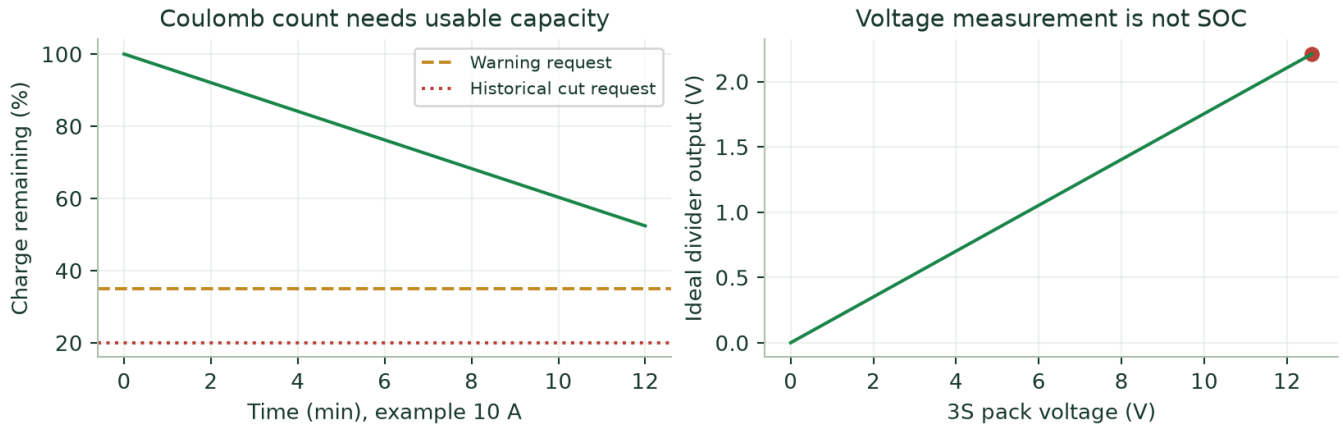
PLANNED CAMPAIGN + STATISTICAL LAW — No cells represent completed missions. Independence and comparable trials are assumptions, not guarantees.

Test contract: stimulate normal handheld/radio or companion interfaces; log source hashes, messages, estimates and motor requests. Do not apply hidden pose, speed or force corrections to complete a mission. Host-compiled RP2350 C control code requires disclosed shims; it is not RP2350 instruction emulation or hardware timing evidence.

Planned Gazebo campaign: ten environments, five noise seeds each; five additional manual command scenarios. Preserve all failures. Video reviewed at 10-15 fps supports trajectory analysis; full-rate logs establish controller timing. No new campaign has been run for this report.

P targets: attitude RMS <=2 degrees, altitude RMS <=0.20 m, endpoint error <=0.5 m, zero collision in the declared campaign. They are reviewable objectives, not achievements. Zero failures in 50 independent comparable trials gives a 95 percent upper failure probability of about 5.8 percent, not zero risk.

Battery estimation and state policy



ANALYTIC EXAMPLE — This is not measured endurance. Whole-pack monitoring cannot protect each cell; a motor cut in flight causes a fall.

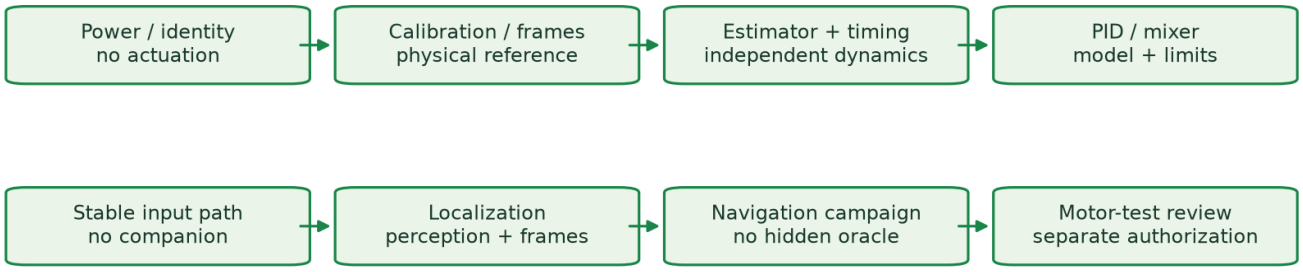
$$SOC(t) = SOC(0) - \frac{1}{C_{usable}} \int_0^t I(\tau)\, d\tau, \qquad V_{ADC} = V_{pack} \frac{R_{bottom}}{R_{top} + R_{bottom}}$$

Measurements: calibrated whole-pack voltage and current, useful capacity, voltage sag and sensor validity. The plots show ideal laws, not measured endurance or an SOC calibration. Marked 120C and four 40 A ESCs do not establish a usable total discharge current.

Historical requested thresholds: warning at 35 percent and motor disarm at 20 percent, without automatic safety landing. A motor cut in flight causes a fall; reconfirm this policy explicitly before any flight phase. Hysteresis, persistence, uncertainty and fault responses remain TBD. Whole-pack software monitoring does not protect individual cells.

Current states: BOOT -> SELF_TEST -> BENCH_READY only. Reset does not arm. Physical motor outputs remain disabled. One verified regulated avionics source; common grounds; no parallel BEC positives; propulsion current stays outside FC copper. Separate authorization is required for motor tests.

Project gates risks and personal development



PROPOSED ARCHITECTURE — Offline preparation is allowed; failed earlier gates invalidate affected downstream acceptance.

Risk or dependency	Engineering response
Wrong axes / signs / motor mapping	Known signed rotations and mixer checks before actuators
Lost data / I2C hang / stale commands	Bounded waits, validity/age checks and explicit degraded state
Saturation / boost / mode transition	Headroom, anti-windup and bumpless references
Missing hardware or deadline	Deliver the verified subset; prepare later stages offline without acceptance
Camera range / localization gaps	Restrict envelope; do not infer unobserved free space

Professional outcomes: requirements -> model -> experiment -> interpretation -> decision. The personal reflection should explain one real difficulty, the student's action, the evidence and the change in understanding. Do not invent experience or sole authorship. Disclose library reuse and AI assistance under institutional policy.

Management: commit each meaningful change; preserve failed evidence; invalidate affected downstream acceptance after an earlier-stage failure. No deadline or rubric was supplied, so the schedule is gate-driven rather than an invented calendar. Add authentic contributions, cover fields, quotations and the actual rubric before submission. Reuse modular hardware and tools; follow appropriate battery, electronic-waste and etching-chemical procedures.

Materials and resources

Installed aircraft BOM: quantities refer to one intended aircraft, not confirmed stock. Verify exact models, physical dimensions, ratings, ownership and masses. Prices are blank until dated quotations are obtained. Optional/future hardware is listed separately on the next page.

ID / qty	Material	Purpose / acceptance
M01 x1	F450 or Q450 X frame with plates and arms	Structural assembly; verify motor diagonal, stiffness and mounting holes
M02 x4	Nominal 1400 KV brushless motors	Exact model TBD; inspect shaft, mounts and propeller compatibility
M03 x4	Nominal 40 A ESCs	Verify voltage range, input protocol and which unit has a suitable BEC
M04 x4	9047 propellers, matched CW and CCW pairs	Verify handedness, hub fit, balance and motor recommendation
M05 x1	3S 4200 mAh LiPo, marked 120C	Inspect condition; weigh; characterize useful capacity and current demand
M06 x1	RP2350 development board, variant TBD	Pico SDK FC; verify board, memory, interfaces and 1 kHz timing
M07 x1	GY-521 MPU6050 eight-pin module	Check device identity, pin order, offsets and mounting orientation
M08 x1	BMP280 six-pin module	Confirm BMP rather than BME identity; verify I2C at 0x76
M09 x1	GY-271 style magnetometer module	Identify actual HMC/QMC silicon and matching driver; inspect 3.3 V pull-ups
M10 x1	TOFSense F2 Mini and cable	Verify exact model, saved I2C mode, connector orientation and range status
M12 x1 set	Power distribution assembly	Rated wiring/connectors; common reference; propulsion current outside FC
M13 x1	Regulated FC and sensor supply	One verified source; chosen RP2350 board input range and load checked
M14 x1 set	RP2350 carrier PCB and module sockets	User-made; target pin assignment, fit and electrical checks pending
M15 x1 set	Fasteners, spacers, battery straps and insulation	Secure battery and rigid sensor reference; prevent conductive contact
M16 x1 set	Battery divider and ADC RC filter	Candidate 47 kOhm / 10 kOhm, 1 percent resistors and 100 nF; verify ADC range
M17 x1	Current measurement assembly	Sensor/range TBD; compatible conditioned output and external high-current path
M18 x4	Three-pin ESC signal headers	Label GND, isolated BEC positive and signal; cable order must be checked
M19 x1 set	Lidar solder pads or connector	Four conductors; strain relief for remote sensor

Selection logic: RP2350 is the aircraft FC and RP2040 the custom handheld. One LoRa and two reserved CC1101 modules are allocated to each side. The exact radio modules, display, antennas and handheld power remain to choose. The magnetometer's actual silicon and driver must be identified rather than trusting the GY-271 board name. The ESP32/Xbox branch is historical experimental work, not the target bill of materials.

Measurement equipment optional hardware and cost

ID / qty	Material	Purpose / acceptance
M11 x1	RP2040 development board	Custom handheld; joystick, trigger, button, display and radio interfaces
T01 x1	USB-TTL adapter	3.3 V logic, 921600 baud capability and suitable 5 V lidar power
T02 x1	Digital multimeter	Voltage, resistance and continuity; known uncertainty
T03 x1	Current instrument	Suitable DC clamp or rated inline sensor; never place ammeter across battery
T04 x1 set	Scale, level, angle fixture and ruler	Record resolution and repeatability; rigid fixtures preferred to hand-held poses
T05 x1	3S balance charger and suitable charging setup	Use battery-compatible settings and inspect pack condition
T06 x1 set	Soldering, insulation and inspection tools	Assembly quality, magnification and continuity before power
O01 x1	Raspberry Pi 4/5 OR NVIDIA Jetson, plus storage	Sought, not selected; benchmark latency, power, mass, OS and cost
O02 x1	Compatible RGB camera and mount	Optional perception; calibrate intrinsics, extrinsics and timestamps
O03 x1 set	Companion power and serial interface	Optional; size separately and verify logic level and shared reference
M20 x2	Centred two-axis joystick modules	Four analogue axes; neutral, endpoint and noise calibration
M21 x1 set	LB/RB, direction and mode buttons	Final arrangement and gesture mapping TBD; debounce required
M22 x2	LT/RT trigger controls	Analogue or digital choice to confirm; calibrate released/pressed states
M23 x1	74HC4051 analogue multiplexer	Existing RP2040 prototype acquisition option; settling/crosstalk checks
M24 x1	Handheld display, model TBD	Readable link, mode, battery and status information
M25 x1 set	Handheld power and enclosure	Battery/regulation/charging and mechanical design TBD
R01 x2	LoRa modules, exact type and band TBD	One per side, primary pilot link; airtime and loss tests
R02 x4	CC1101 modules	Two per side, reserved secondary links; not implemented fallback
R03 x6	Matched antennas and RF connections	Match selected modules/bands; isolation and coexistence checks

$$C_{base} = \sum_i n_i c_i + C_{shipping} + C_{fabrication}, \qquad C_{planned} = C_{base} + C_{contingency}$$

Use the CSV inventory for owned/to-buy status, vendor, currency, quote date, unit cost and measured mass. Separate shared tools from aircraft cost. A 15 percent contingency with 10/25 percent sensitivity is only a planning scenario. Total cost remains TBD; no supplier price is fabricated.

Verification matrix and proposed acceptance

Test	Coverage	Essential evidence
V01	IF-01, FC-09, COM-01	Power, ground, non-actuating outputs, companion absence
V02	EST-02, IF-02	Cold boot, reset, MPU recovery and readback
V03	EST-01/03/04, MOD-01	Independent poses, signed rotations, uncertainty
V04	IF-03/04, EST-06/07	Known range, invalid returns, removal/restart
V05	RT-01/02/03/04/06	Full-load timing, queue/loss counters and deadlines
V06	FC-02/03/05/07/08, MOD-02/03/04	Model, gains, steps, saturation, source path
V07	HMI-01/02/03/04, FC-01/06, SYS-02	RP2040 inputs, mapping, gestures and command generation
V08	COM-02/03, EST-06, RT-04	CRC, replay, stale/delayed messages
V09	FC-04, EST-05	Vertical response and lost sensor
V10	NAV-01/02/05	All mission nodes, frames and localization
V11	NAV-03/04	Off-ray/unclassified obstacles; no oracle
V12	ENE-01/02/03	Reference power measurement and threshold faults
V13	RF-01/02	LoRa airtime/age/loss/reconnect and CC1101 coexistence; reserved links remain inactive

Every result: commit and source/configuration hashes, engine or hardware identity, original logs, assumptions, fixed criteria, verdict and limits. New gains or configuration invalidate incompatible acceptance. An attractive video is insufficient without the command path and data. No failure is deleted to create a pass.

Assessment evidence: mathematical reasoning, sourced design choices, reproducible original measurements, adverse-result interpretation, resource/risk management and authentic reflection. No guaranteed mark or MIT endorsement is claimed.

Requirements register one

The historical baseline is preserved separately. This register retains the original identifiers, updates platform-specific handheld/software requirements and adds target-system/radio requirements. RP2040 gesture mapping remains TBD rather than silently inheriting Xbox A/B buttons. Detailed pinout/API syntax stays outside the main submission.

ID / status	Requirement	Verify
FC-01 D	Standalone stable mode without companion	V01,V07
FC-02 D	Yaw angular-rate control	V06
FC-03 D	Roll/pitch angle loops with inner rate loops	V06
FC-04 D	Vertical speed input; neutral height hold	V09
FC-05 D	No initial horizontal braking or position hold	V06
FC-06 D	Gentle bounded input shaping	V07
FC-07 P	Boost headroom; collective ceiling 1700-1800 us TBD	V06
FC-08 D	Saturation and integral anti-windup	V06
FC-09 D	Current physical arming and motor outputs disabled	V01
HMI-01 D	RP2040 centred joysticks, LB/RB, LT/RT and mode buttons; Mode 2	V07
HMI-02 D-mapTBD	Stable-mode request; final RP2040 button gesture TBD	V07
HMI-03 D-mapTBD	Ready navigation request; final RP2040 button gesture TBD	V07,V10
HMI-04 D-mapTBD	Explicit disarm request; final handheld gesture TBD	V07
EST-01 D	Calibration tied to mounting, frames and held-out data	V03
EST-02 D	Boot trim restore/readback; no double bias correction	V02
EST-03 D	Gyro bias, scale and physical frame verification	V03
EST-04 D	Angle estimation noise and acceleration limits declared	V03,V06
EST-05 D	Vertical estimator with z and vz states	V09
EST-06 D	Detect stale, repeated, invalid and saturated samples	V04,V08
EST-07 P	Slow observations retain timestamp and age	V04,V05
RT-01 D	Complete FC target 1000 Hz	V05
RT-02 P	Critical-work budget at most 800 us	V05
RT-03 P	No observed miss in defined 10 min loaded test	V05
RT-04 D	Bounded waits, buffers, communication and logging	V05,V08
RT-05 D	Pico SDK single C runtime per FC/handheld; normal ROS C++ packages	Inspect/build

Requirements register two

ID / status	Requirement	Verify
RT-06 D	Distinguish sample, estimation, PID and ESC rates	V05
IF-01 D	3.3 V I2C logic; verified power and common ground	V01
IF-02 D	MPU bypass restore/readback after reset	V02
IF-03 D	Lidar I2C configuration/address verified	V04
IF-04 P	Slow bus starts 100 kHz; changes retested	V04,V05
COM-01 D	Companion optional; no automatic mode change	V01,V07
COM-02 P	Framed/versioned UART, CRC, sequence and validity	V08
COM-03 P	Heartbeat 10 Hz and expiry 300 ms pending review	V08
NAV-01 D	Visit every approved node in order	V10
NAV-02 P	0.5 m radius, 1 s dwell and speed criterion TBD	V10
NAV-03 D	Replan from sensed obstacles; no hidden truth	V11
NAV-04 D	Separate detections, range and observed free space	V11
NAV-05 TBD	Select and validate observable metric localization	V10,V11
MOD-01 D	Explicit coherent world/body/sensor frames	V03
MOD-02 D	Physical/LTI derivations and verified discretization	V06
MOD-03 D	Identified gains, stability margins and saturation tests	V06
MOD-04 D	Separate parameter uncertainty, noise, bias and delays	V06
ENE-01 D	Calibrated whole-pack voltage/current measurement	V12
ENE-02 D	Charge estimate distinguishes capacity, sag and integration	V12
ENE-03 D-review	Historical 35% warning / 20% cut request; reconfirm before flight	V12
SYS-01 D	RP2350 aircraft FC target; 1 kHz acceptance on actual hardware	V05
SYS-02 D	RP2040 custom handheld with display and calibrated inputs	V07
SYS-03 TBD	Seek/select Raspberry Pi or Jetson companion by measured constraints	Review
RF-01 D	LoRa primary pilot link with bounded command age	V13
RF-02 D	Two reserved CC1101 links per side; no assumed redundancy	V13

Software inventory: Pico SDK and C toolchain; one flight_controller.c and one handler_controller.c; selected radio/sensor libraries; ROS 2 C++; Gazebo physics; RViz; YOLO inference; NAssistant configuration; Python analysis; KiCad; Git/GitHub. ESP32/Arduino/Adafruit/Xbox dependencies belong to preliminary evidence only. Pin versions, licences, model hashes and commands. Isaac Sim is excluded.

Evidence access: original private recordings must be made available to the examiner through an appropriate sanitized evidence package. Public summaries/hashes are not substitutes for the originals. The supplied model parameters are not independently measured aircraft properties.

References and evidence provenance

Primary technical sources: [R1] NASA systems-engineering verification/validation: https://www.nasa.gov/reference/2-0-fundamentals-of-systems-engineering/ . [R2] Espressif ESP32-S3 datasheet and BLE support: https://www.espressif.com/sites/default/files/documentation/esp32-s3_datasheet_en.pdf . [R3] TDK MPU6050 register map: https://invensense.tdk.com/wp-content/uploads/2015/02/MPU-6000-Register-Map1.pdf . [R4] Bosch BMP280 datasheet: https://www.bosch-sensortec.com/media/boschsensortec/downloads/datasheets/bst-bmp280-ds001.pdf .

[R5] Nooploop TOFSense-F V2.0, hardware and protocol: https://support.nooploop.com/tofsense/ . [R6] Historical Xbox BLE prototype library: https://github.com/asukiaaa/arduino-XboxSeriesXControllerESP32 . [R7] Ultralytics detector outputs: https://docs.ultralytics.com/tasks/detect/ . [R8] MIT OCW 16.30 feedback control: https://ocw.mit.edu/courses/16-30-feedback-control-systems-fall-2010/ . [R9] ROS frame conventions: https://www.ros.org/reps/rep-0105.html . Reviewed 28 October 2026; sources support methods, not acceptance of this aircraft.

[R10] Raspberry Pi RP2350 and RP2040 specifications: https://www.raspberrypi.com/products/rp2350/ and https://www.raspberrypi.com/documentation/microcontrollers/microcontroller-chips.html . [R11] TI CC1101 transceiver: https://www.ti.com/product/CC1101 . [R12] Semtech LoRa airtime trade-off: https://www.semtech.com/design-support/faq/faq-lora . [R13] ROS 2 action and QoS contracts: https://github.com/ros2/ros2_documentation/blob/jazzy/source/Concepts/Basic/About-Actions.rst and https://github.com/ros2/ros2_documentation/blob/jazzy/source/Concepts/Intermediate/About-Quality-of-Service-Settings.rst .

Implementation evidence and report provenance

Project evidence: [P1] measurements/mpu6050/2026-10-25-secured-mount-analysis-v2/report.md and analysis.json; [P2] measurements/mpu6050/2026-10-26-gyro-restart-validation/report.md; [P3] measurements/mpu6050/2026-10-26-fifo-and-gyro-frame-results/report.md; [P4] measurements/mpu6050/2026-10-26-offline-attitude-preparation/report.md and analysis.json; [P5] measurements/software/2026-10-27-input-guard/summary.json. Paths are relative to the Stepwise checkout. Evidence-manifest.json records source hashes and repository links.

[P6] Inspected prototype code: main commit bce665ad97adbfcb282f02970a1db599e67e6479, firmware/flight_controller/flight_controller.c, firmware/handler_controller/handler_controller.c and companion/ros2_ws/src/qcopter_autonomy. Read-only source inspection supports the implementation inventory; no new Pico build, upload, radio test or ROS flight is claimed. source-inventory.json records exact file hashes and repository links.

Figure provenance: assets/visual/figure-manifest.json identifies every chart as a historical measured summary, replay/model result, analytic example, schematic or proposal. Model-only plots use explicit assumptions and do not represent Gazebo or hardware tests. The F450 artwork is AI-generated using the built-in image tool; image-provenance.json retains its prompt and origin. It depicts a plausible red/white F450, not the student's assembled hardware.

Preserved detail: reference/engineering-baseline-v2.1.md and reference/visual-edition-v3.md retain the earlier ESP32-oriented documents. Their pin maps and Xbox-specific gestures are historical, not the Pico target interface. This revision updates the intended architecture and explanatory design; it neither enables motors nor accepts RP2350 timing, radio delivery or autonomous flight. No firmware or calibration constant is changed.